

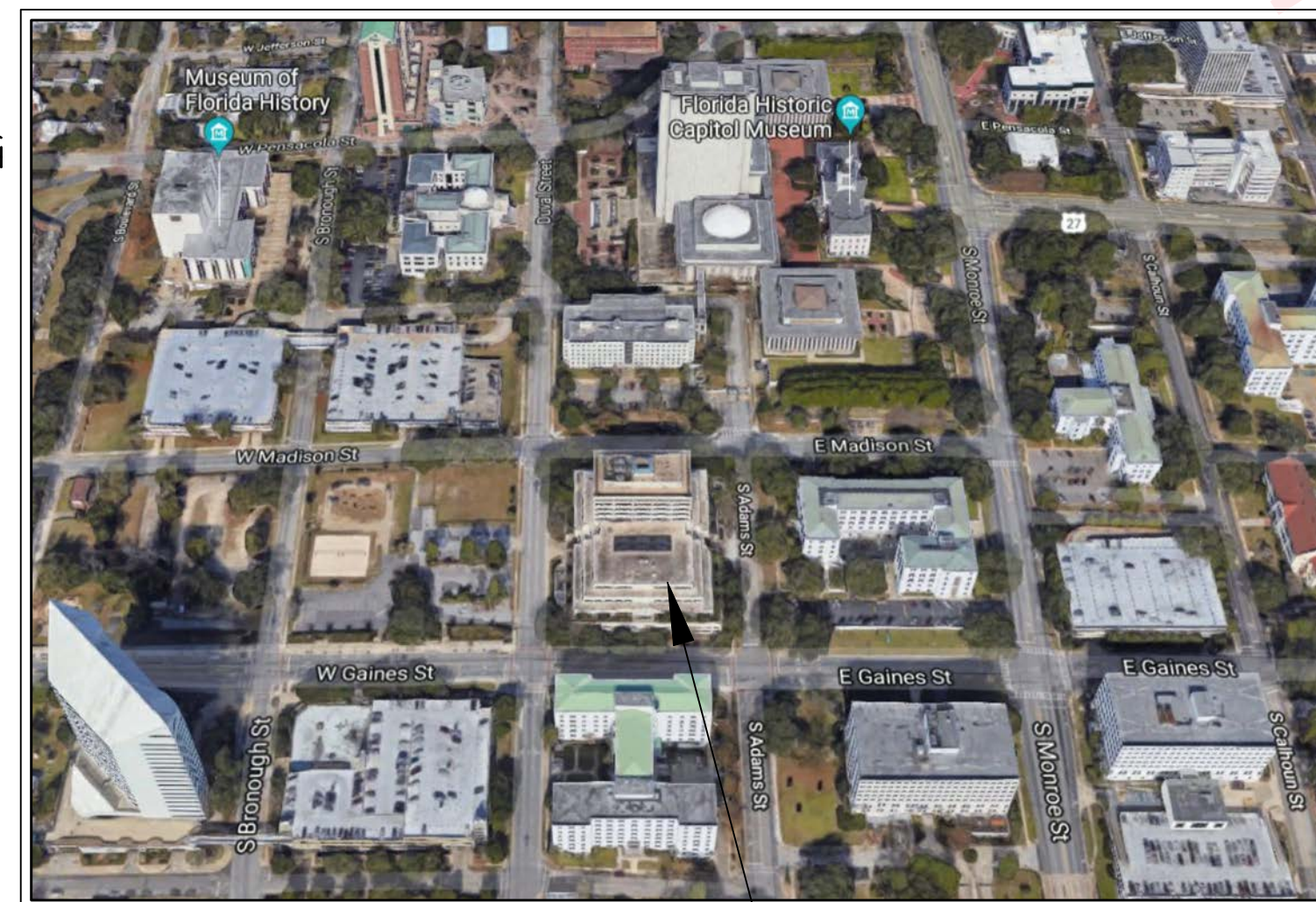


MSFM 02203280

DESIGN REPLACEMENT OF MAJOR ELECTRICAL SYSTEMS - PHASE 2 PEPPER BUILDING

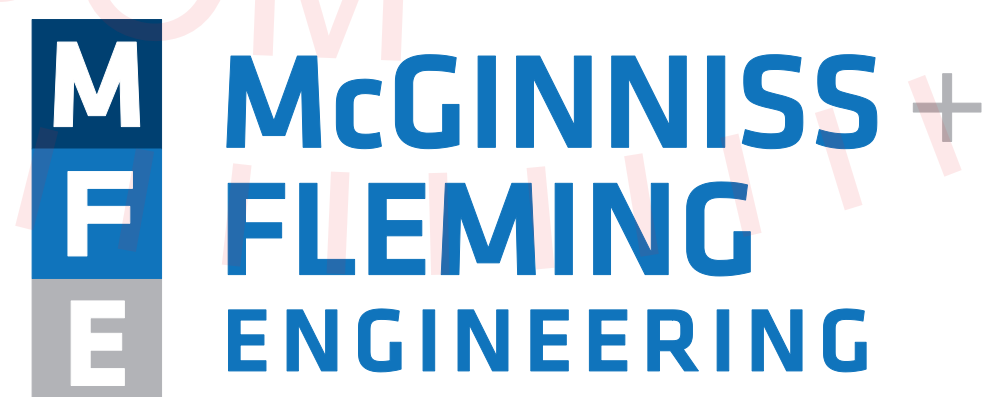
TALLAHASSEE, FLORIDA

Area Map



CLAUDE PEPPER BUILDING

100% CONSTRUCTION DOCUMENTS
April 30, 2024



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SYMBOL LEGEND

ABBREVIATIONS

27	UNDervOLTAGE RELAY
43	AUTO/MANUAL TRANSFER DEVICE
50	INSTANTANEOUS OVERCURRENT RELAY
50AF/50GAF	ARC FLASH OVERCURRENT
50BF	BREAKER FAILURE
50/51G	GROUND FAULT PROTECTION
51	AC TIME OVERCURRENT RELAY
52	AC 15KV CIRCUIT BREAKER
59	OVERVOLTAGE RELAY
68	BLOCKING RELAY
81	FREQUENCY OVER/UNDER RELAY
83	AUTOMATIC TRANSFER RELAY
86	LOCK-OUT RELAY
87	DIFFERENTIAL PROTECTIVE RELAY (HIGH IMPEDANCE)
A	AMPERE
AC	ABOVE CEILING, ABOVE COUNTERTOP, ALTERNATING CURRENT
AFC	AVAILABLE FAULT CURRENT
AFF	MOUNTING HEIGHT ABOVE FINISHED FLOOR OR GRADE TO CENTERLINE
AIC	EQUIPMENT AMPERE INTERRUPTING CAPACITY
AM	AMMETER
AS	AMMETER SELECTOR OPERATOR
BFC	BELOW FINISHED CEILING
CKT	CIRCUIT
COT	CITY OF TALLAHASSEE
CPT	CONTROL POWER TRANSFORMER
CS	CONTROL SWITCH
CT	CURRENT TRANSFORMER
DC	DIRECT CURRENT
DMS	DEPARTMENT OF MANAGEMENT SERVICES
EC	EMPTY CONDUIT (3/4" MINIMUM) WITH NYLON PULLWIRE
ECB	ENCLOSED CIRCUIT BREAKER
EM	EMERGENCY
EX	EXISTING - RECONNECT AS REQUIRED AT EXISTING LOCATION. REMOVE AND REINSTALL IF REQUIRED
ETR	EXISTING TO REMAIN
GEMB	GROUNDING ELECTRODE MAIN BUSBAR
GFI	GROUND FAULT INTERRUPTER
J	JUNCTION
KW	KILOWATT METER
KWM	KILOWATT-HOUR METER
LOTO	LOCK OUT / TAG OUT
PT	POTENTIAL TRANSFORMER
R	RELAY
SCADA	SUPERVISORY CONTROL AND DATA ACQUISITION
T	TRANSFORMER
TBR	TO BE REMOVED
UNO	UNLESS NOTED OTHERWISE
V	VOLT
VA	VOLT-AMPS
VM	VOLTMETER
VS	VOLTMETER SELECTOR OPERATOR
W	WATT
WP	WEATHERPROOF (NEMA 3R)

HOMERUNS TO PANELS

	ARROW INDICATES CIRCUIT HOMERUNS IN CONDUIT
	INDICATES HOMERUN TO CIRCUIT NUMBERS 2 & 4 IN PANEL "LPA"
NOTE: NUMBER OF HOMERUNS SHOWN ON THE PLANS ARE THE NUMBER OF HOMERUNS REQUIRED. DO NOT RUN MORE THAN THREE HOMERUNS IN ONE CONDUIT. DO NOT RUN 2 CIRCUITS ON THE SAME PHASE IN ONE CONDUIT.	
	INDICATES CONTINUATION OF RUN SHOWN ON ANOTHER PLAN VIEW

CIRCUITING AND BRANCH CIRCUITS

	INDICATES 1 #12 PHASE CONDUCTOR, 1 #12 NEUTRAL & 1 #12 GND - 3/4" C. TO 1-POLE BREAKER, SIZED AS SHOWN IN PANEL SCHEDULES, ON CIRCUIT No. 3, IN PANEL 'L2A'.
	INDICATES 2 #12 PHASE CONDUCTORS & 1 #12 GND - 3/4" C. TO 2-POLE BREAKER, SIZED AS SHOWN IN PANEL SCHEDULES, ON CIRCUIT No.'s 2 & 4 IN PANEL 'L2A'.
	INDICATES 3 #12 PHASE CONDUCTORS & 1 #12 GND - 3/4" C. TO 3-POLE BREAKER, SIZED AS SHOWN IN PANEL SCHEDULES, ON CIRCUIT No.'s 2,4,6 IN PANEL 'L2A'.
	INDICATES ALL CONDUCTORS ARE TO BE MINIMUM #10 GAUGE, CONDUIT PER NEC OR AS INDICATED.
	SHORTER TICKMARKS INDICATE 2 OR MORE PHASE CONDUCTORS, OR SWITCH LEGS
	LONGER TICKMARKS INDICATE GROUNDED CONDUCTOR(S), QUANTITY AS SHOWN.
NEUTRALS SHALL NOT BE SMALLER SIZE THAN PHASE CONDUCTORS UNLESS SPECIFICALLY INDICATED OTHERWISE. PROVIDE THE APPROPRIATE NUMBER OF NEUTRALS IN ACCORDANCE WITH NEC.	
INSULATED GROUNDING CONDUCTORS SHALL BE USED IN ALL CIRCUITS, SIZED IN ACCORDANCE WITH NEC ARTICLE 250.	
2-#12, 1-#12 GROUND SHALL BE RUN IN 3/4" CONDUIT. 4 OR MORE #12 CONDUCTORS SHALL BE RUN IN 3/4" C. OR AS REQUIRED BY NEC. LARGER THAN #12 CONDUCTORS SHALL BE RUN IN CONDUIT SIZED IN ACCORDANCE WITH NEC.	
	CONCEALED OVERHEAD OR IN WALLS
	CONCEALED BELOW FLOORS OR GRADE

POWER PANELS

	PANELBOARD 208 OR 480 VOLT
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LIGHTING OUTLETS

	GEAR CONTROLLED LIGHTING OUTLET TO BE REMOVED
	LIGHTING FIXTURE - MOUNTING AND TYPE AS SHOWN IN FIXTURE SCHEDULE. SEE SCHEDULE FOR SPECIFIC REQUIREMENTS.

JUNCTION BOXES

	JUNCTION BOX
--	--------------

LIGHTING CONTROLS

	FLUSH TYPE, 20A, 120/277V AC ONLY, QUIET TYPE, SINGLE POLE SWITCH
	FLUSH TYPE, 20A, 120/277V AC ONLY, QUIET TYPE, 3 WAY SWITCH

CONTROLS & MECHANICAL EQUIPMENT

	MOTOR CONTROLLER - MAGNETIC - MOUNTS IN SEPARATE ENCLOSURE FURNISHED AND INSTALLED BY DIVISION 16 - COORDINATE LOCATION WITH MECHANICAL STARTER SIZE AS INDICATED OR AS REQUIRED FOR MOTOR SERVED
	MOTOR SNAP SWITCH WITH THERMAL OVERLOAD PROTECTION - 20A, 1 POLE UNLESS NOTED OTHERWISE.
	3/4" X 10'-0" GROUND ROD.
	GROUND TO EQUIPMENT
	PILOT LIGHT - RED
	PILOT LIGHT - AMBER
	PILOT LIGHT - GREEN
	PILOT LIGHT - WHITE
	GROUND FAULT PROTECTION MONITORING
	SURGE PROTECTION DEVICE
	ENERGY REDUCING MAINTENANCE SWITCH WITH LOCAL STATUS INDICATOR

SWITCHGEAR AND ACCESSORIES

	BREAKER CONTROL SWITCH
	CONTROL RELAY OR METER, SEE DESCRIPTION
	FUSE
	CURRENT TRANSFORMER
	VOLTAGE TRANSFORMER
	PROTECTIVE RELAY, SEE DESCRIPTION
	SEPARABLE CABLE CONNECTOR OR EQUIPMENT CONNECTION
	CIRCUIT IN CONDUIT CONCEALED
	CIRCUIT IN CONDUIT - CONTINUATION
	CIRCUIT ENTRY AT ENCLOSURE (UP)
	CIRCUIT ENTRY AT ENCLOSURE (DOWN)
	CIRCUIT BREAKER



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PEPPER BUILDING
DESIGN REPLACEMENT OF
MAJOR ELECTRICAL SYSTEMS
PHASE 2

MSFM-02203280

Tallahassee, Florida

DATE:
April 30, 2024

REVISED:

DESIGNED BY:
BW

DRAWN BY:
TB

SUBMITTAL:
100% Construction Documents

SHEET TITLE:

SYMBOL LEGEND

SHEET:

E100

JOB NUMBER:

2311

GENERAL NOTES

1. PHASING SHALL BE COORDINATED AND IN COMPLIANCE WITH ALL PHASING DRAWINGS AND NOTES.
2. ALL CONDUCTORS SHALL BE INSTALLED IN METAL CONDUIT OR TUBING. FLEXIBLE CONDUIT INSTALLED OUT-OF-DOORS, IN ANY MECHANICAL EQUIPMENT ROOM, OR IN NORMALLY WET AREAS, SHALL BE LIQUID TIGHT FLEX WITH UL LISTED FITTINGS.
3. CONDUIT SHALL BE INSTALLED TIGHT TO DECK WHERE INSTALLED ABOVE CEILING. RELOCATE INCIDENTAL HANGERS, BRACKETS, ETC. WHERE IN CONFLICT. MAXIMIZE USE OF SPACE.
4. CONDUIT SHALL PASS THROUGH WALLS AT 90 DEGREES AND SHALL BE RUN PARALLEL AND PERPENDICULAR TO WALLS.
5. ALL CONDUCTORS SHALL BE COPPER WITH THHN/THWN INSULATION. CONDUCTORS SIZE # 8 AWG AND LARGER SHALL BE STRANDED.
6. NO MORE THAN 3 PHASE CONDUCTORS SHALL BE INSTALLED IN ONE CONDUIT UNLESS NOTED OTHERWISE.
7. MAINTAIN NEC MINIMUM CLEARANCE IN FRONT OF ALL SAFETY SWITCHES, PANELBOARDS, AND SWITCHGEAR.
8. ALL UNDERGROUND CONDUIT RUNS ENTERING THE BUILDING SHALL BE SEALED TO PREVENT THE ENTRANCE OF MOISTURE AND GASES.
9. ALL ITEMS ON PLANS ARE NEW UNLESS NOTED OTHERWISE.
10. ALL WALL PENETRATIONS SHALL BE SEALED USING AN APPROPRIATE UL ASSEMBLY TO MAINTAIN THE RATING OF THE WALL.
11. ALL SINGLE POLE CIRCUITS SHALL TERMINATE ON A SINGLE POLE CIRCUIT BREAKER ASSEMBLY. DO NOT USE MULTI-POLE CIRCUIT BREAKER OVERCURRENT PROVISIONS FOR 120 VOLT OR 277 VOLT CIRCUITS. PROVIDE INDIVIDUAL GROUNDED CONDUCTORS FOR ALL SINGLE POLE CIRCUITS.
12. EXISTING PANELS TO REMAIN SHALL HAVE THEIR PANEL SCHEDULES UPDATED AND RETYPED AFTER COMPLETION OF NEW WORK. NEW PANEL LOAD DESCRIPTIONS SHALL CLEARLY DEFINE LOAD TYPE AND LOCATION. PANEL SCHEDULES, BOTH FOR NEW AND EXISTING PANELS, SHALL INCLUDE CONTRACTOR COMPANY NAME, PHONE NUMBER, AND DATE SCHEDULE WAS MADE AND APPLIED.
13. ALL GROUND FAULT PROTECTION DEVICES INSTALLED ON SERVICE DISCONNECTING MEANS, WHERE REQUIRED, SHALL BE PERFORMANCE TESTED WHEN FIRST INSTALLED ON SITE. TESTING SHALL BE CONDUCTED BY A QUALIFIED PERSON USING A TEST PROCESS OF PRIMARY CURRENT INJECTION AND IN ACCORDANCE WITH MANUFACTURE'S INSTRUCTIONS. TESTING REPORT SHALL BE MADE AVAILABLE TO THE AHJ AND THE OWNER.
14. WHERE CORE DRILLING OCCURS, CONTRACTOR SHALL USE CAUTION TO AVOID DAMAGING SURROUNDING SURFACES. CONTRACTOR RESPONSIBLE FOR CLEANING UP ALL SLURRY AND ENSURING A CLEAN AND PROFESSIONAL FINISH. CONTRACTOR SHALL REPLACE OR REPAIR BUILDING SURFACES OR EQUIPMENT DAMAGED BY SLURRY OR CORE DRILLING MATERIALS.
15. WHERE INSTALLING ELECTRICAL CONDUIT, MAINTAIN CONDUIT STRAPPING AND FASTENING PER NEC.
16. WHERE INSTALLING ELECTRICAL CONDUIT, MAINTAIN PULL POINTS PER NEC AND DO NOT EXCEED 360 DEGREES IN BENDS IN ANY CONDUIT RUN WITHOUT APPROPRIATE PULL POINTS.
17. ALL JUNCTION BOXES, PULLBOXES, AND DEVICES SHALL BE LABELED WITH CIRCUIT NUMBER, CIRCUIT VOLTAGE, AND FED TO / FROM PANEL LOCATIONS.
18. CONTRACTOR SHALL VERIFY WORKING SPACE AND DEDICATED ELECTRICAL SPACE EXISTS FOR PANELS BEING MODIFIED IN ACCORDANCE WITH NEC. CONTRACTOR SHALL BRING ANY DEFICIENCIES FOUND TO THE ATTENTION OF THE ENGINEER.

SCOPE OF WORK

1. FURNISH ALL LABOR, MATERIALS, EQUIPMENT AND INCIDENTALS REQUIRED TO COMPLETE ALL ELECTRICAL WORK AS SHOWN ON THE CONTRACT DRAWINGS.
2. THIS SHALL INCLUDE THE INSTALLATION OF A COMPLETE AND PROPERLY OPERATING ELECTRICAL SYSTEM. THIS SYSTEM REQUIRED CONSISTS BASICALLY OF, AND IS NOT LIMITED TO, THE FOLLOWING:
 - EXTEND THE DISTRIBUTION SYSTEM FOR POWER INCLUDING THE NECESSARY FEEDERS, BRANCH CIRCUITS, INSTALLATION OF AND CONNECTION TO DEVICES, PANELBOARDS, TRANSFORMERS, SWITCHES, LOW-VOLTAGE SWITCHGEAR, MEDIUM VOLTAGE SWITCHGEAR, AND ALL OTHER EQUIPMENT SHOWN.
 - PERFORM RETROFILL OF THE NOTED 480V SWITCHGEAR AND SUB-SYSTEMS, INCLUDING BUT NOT LIMITED TO, REPLACEMENT OF CIRCUIT BREAKERS, METERING SYSTEM COMPONENTS, CT's, PT's, CPT's, TERMINAL BOARDS, FRONT DOOR PANELS, AND ALL SWITCHGEAR CONTROL WIRING AND CONNECTIONS. COMPLETE CLEANING AND PAINTING OF EXISTING 480V SWITCHGEAR ENCLOSURE PANELS AS NOTED IN PLANS AND SPECS.
 - COMPLETE TESTING OF GROUNDING SYSTEMS AS NOTED IN PLANS AND SPECS.
 - EXISTING INCOMING AND OUTGOING FEEDERS SHALL BE PROTECTED DURING CONSTRUCTION, UNLESS SHOWN TO BE REPLACED. REFER TO PLANS AND ONE LINES FOR FEEDERS TO BE REPLACED AND / OR REWORKED TO ACCOMMODATE SWITCHGEAR RETROFILL.
 - PERFORM SWITCHING PROCEDURES AT 15KV PEPPER SWITCHGEAR FOR ISOLATION OF 480V SWITCHGEAR TO BE RETROFILLED. OUTAGES TO OLTS GEAR ARE NOT PERMITTED. REFER TO SEQUENCING PLANS.
 - PROVIDE NETWORK INFRASTRUCTURE FOR SCADA. PERFORM ALL SCADA WORK AS INDICATED ON SCADA PLAN, INCLUDING PROGRAMMING OF EXISTING PME SYSTEM.
 - PERFORM THE DEMOLITION WORK INDICATED.
 - THE CONTRACTOR IS RESPONSIBLE FOR ALL TEMPORARY WORK, RIGGING, CRANES AND LIFTS, TRAFFIC CONTROL AND OTHER WORK THAT IS NECESSARY TO ACCOMPLISH THE SWITCHGEAR REPLACEMENT IN A SUITABLE MANNER, INCLUDING BUT NOT LIMITED TO, FENCING AND BOLLARD REMOVAL AND PUTBACK.
 - SURFACE FLOOR WITH PLYWOOD OR SIMILAR SUITABLE MATERIAL ALONG COMPLETE TRANSPORTATION PATHWAY TO PREVENT MARRING DURING REMOVAL AND INSTALLATION OF EQUIPMENT. DO NOT ROLL OR SKID LIFTS, HOISTS, JACKS, ETC. ACROSS FINISH FLOOR. PROTECTION IS TO REMAIN FOR DURATION OF INSTALLATION. ADHERE TO ROUTING INDICATED FOR REMOVAL AND INSTALLATION OF EQUIPMENT.
 - PERFORM SCOPE OF WORK AS INDICATED FOR TEMPORARY POWER PROVISIONS DURING OUTAGES.
 - PERFORM SEALING OF ALL WALLS PER UL DETAILS WHERE EXISTING EQUIPMENT IS REMOVED, LEAVING OPENINGS WITHIN WALL, AS WELL AS, WHEN NEW PENETRATIONS ARE MADE TO RATED WALLS.
 - PRINT AND APPLY ARC FLASH LABELS TO EQUIPMENT. ARC FLASH LABELS WILL BE PROVIDED TO CONTRACTOR IN PDF FORMAT.
 - COMPLETE TESTING OF EXISTING EQUIPMENT AS NOTED.
3. THE BIDDER SHALL INSPECT THE PRESENT JOBSITE CONDITIONS BEFORE PREPARING HIS BID. THE SUBMISSION OF A BID WILL BE CONSIDERED EVIDENCE THAT SUCH A VISIT AND INSPECTION WAS PERFORMED BY THE BIDDER AND THAT HE TAKES FULL RESPONSIBILITY FOR ALL FACTORS GOVERNING HIS WORK.
4. THE ELECTRICAL WORK SHALL BE COMPLETE, FULLY OPERATIONAL, AND SUITABLE IN EVERY WAY FOR THE SERVICE REQUIRED. DRAWINGS ARE GENERALLY DIAGRAMMATIC IN NATURE AND DO NOT SHOW ALL DETAILS. DEVICES AND INCIDENTAL MATERIALS NECESSARY TO ACCOMPLISH THEIR INTENT. THEREFORE, IT SHALL BE UNDERSTOOD THAT SUCH DEVICES AND INCIDENTAL MATERIALS REQUIRED SHALL BE FURNISHED AT NO COST TO THE OWNER.

DEMOLITION NOTES - GENERAL

1. THE LOCATIONS OF ALL ELECTRICAL EQUIPMENT INDICATED (FIXTURES AND DEVICES) MAY VARY FROM DRAWINGS. CONDITIONS AND DEMOLITION WORK WAS DETERMINED BY SITE OBSERVATION AND REVIEW OF EXISTING DOCUMENTS WITHOUT THE BENEFIT OF DESTRUCTIVE INVESTIGATION. VERIFY ACTUAL LOCATIONS, TYPES, CIRCUITING, AND QUANTITIES OF EQUIPMENT AND APPLY DEMOLITION NOTES AS APPROPRIATE FOR THE EQUIPMENT AND ROOM OR AREA.
2. ALL ELECTRICAL ITEMS SHOWN ON DEMOLITION PLANS ARE TO BE REMOVED UNLESS NOTED OTHERWISE. REMOVE ALL ELECTRICAL TRADE MATERIALS ASSOCIATED WITH ITEMS SHOWN TO BE REMOVED. THIS SHALL INCLUDE BUT NOT BE LIMITED TO CONDUIT AND RACEWAYS (BOTH SURFACE MOUNTED AND CONCEALED ACCESSIBLE), WIRE, DEVICES AND CONDUIT SUPPORTS, BACK TO SERVING PANEL.
3. REMOVE ALL CONDUIT STUB-UPS IN FLOOR. CUT OR GRIND CONDUITS BELOW FLOOR LEVEL AND GROUT OPENING. GROUT FINISH SHALL BE ABSOLUTELY FLAT AND READY FOR FLOOR FINISH.
4. ALL REMOVED ELECTRICAL EQUIPMENT, INCLUDING FIXTURES, SWITCHGEAR, BATTERIES, RELAYS, DISCONNECTS, MOTORS, ELECTRICAL CONDUIT, WIRE, DEVICES, AND ALL OTHER MATERIALS SHALL BE DISPOSED OF OFFSITE BY THE CONTRACTOR AT NO ADDITIONAL COST, UNLESS SPECIFICALLY NOTED OTHERWISE.
5. BLANKING PLATES SHALL MATCH NEW PLATES INSTALLED.
6. WHERE EXISTING CIRCUITS ARE REWORKED BY THE ADDITION OR REMOVAL OF CONDUCTORS, THE OLD WIRE SHALL BE REMOVED, THE CONDUIT SWABBED OUT, AND NEW THWN WIRES REPULLED UNLESS INDICATED OTHERWISE.
7. IF ANY EXISTING ELECTRICAL EQUIPMENT THAT IS TO REMAIN BECOMES ISOLATED BY THE REMOVAL OF WALLS, CEILINGS, ELECTRICAL EQUIPMENT, ETC., IT SHALL BE THE CONTRACTOR'S RESPONSIBILITY TO REROUTE AND RECONNECT THE EQUIPMENT SO THAT THE SYSTEM REMAINS OPERABLE. ANY REQUIRED WORK SHALL BE IN ACCORDANCE WITH SPECIFICATIONS.
8. RELOCATE EXISTING ELECTRICAL EQUIPMENT AS REQUIRED TO AVOID NEW CONSTRUCTION. ALL WORK REQUIRED SHALL BE PERFORMED IN ACCORDANCE WITH THE SPECIFICATIONS. PROJECT MANAGER SHALL BE NOTIFIED BEFORE PERFORMING WORK.
9. WHILE REMOVING ALL ELECTRICAL ITEMS INSIDE THE INDICATED ALTERATION AREA, RECONNECT CIRCUITS TO MAINTAIN INTEGRITY OF EXISTING CIRCUITS AND CONTINUED OPERATION OF LIGHTING, EQUIPMENT AND DEVICES TO REMAIN.
10. THE CONTRACTOR SHALL BE RESPONSIBLE FOR ANY CUTTING OR PATCHING OF WALLS AND CEILINGS REQUIRED. CUTTING AND PATCHING WORK SHALL BE PERFORMED BY SKILLED CRAFTSMEN. RETURN ALL WALLS AND THEIR FINISHES TO ORIGINAL CONDITION.
11. ABANDONED CONDUIT SHALL BE REMOVED WHERE POSSIBLE. ABANDONED CONDUIT IN SLAB OR CONCRETE WALLS WHICH CANNOT BE REMOVED SHALL HAVE WIRES PULLED FROM THEM, CUT OFF CONDUIT FLUSH WITH CONCRETE. PATCH OPENINGS AND PENETRATIONS WITH GROUT AND GRIND OR TROWEL SMOOTH, READY FOR NEW FINISH.
12. ANY AND ALL HOLES OR DAMAGED SURFACES DUE TO DEMOLITION WORK SHALL BE PATCHED BY A SKILLED CRAFTSMAN.
13. WHERE ITEMS ARE DESIGNATED EXISTING TO REMAIN, THEY SHALL BE PROTECTED DURING DEMOLITION.
14. ANY HAZARDOUS MATERIAL OR MATERIAL POSING A THREAT TO HEALTH OR THE ENVIRONMENT SHALL BE CAREFULLY DISPOSED OF IN AN APPROVED MANNER. THIS INCLUDES, BUT IS NOT LIMITED TO, FUEL OIL AND CONTAINERS, LIGHT FIXTURE BALLASTS, BATTERIES, BATTERY ACID, OILS, GREASE OR OTHER PETROLEUM PRODUCTS, OR ANTIFREEZE COOLANT.

480V SWITCHGEAR AUTOMATIC TRANSFER SYSTEM

AN AUTOMATIC TRANSFER SCHEME SHALL BE PROVIDED WITHIN THE NEW 480V SWGR FOR THE CONTROL OF THE (2) MAIN CIRCUIT BREAKERS (MAIN 'A' AND MAIN 'B') AND (1) TIE CIRCUIT BREAKER TO TRANSFER THE SECONDARY LOAD SHOULD ONE OR BOTH INCOMING SUPPLIES FAIL. TRANSFER SCHEME SHALL BE SIMILAR TO THAT BELOW.

AUTOMATIC SWITCHING AT NEW 480V SWITCHGEAR UPON LOSS OF INCOMING POWER SHALL INCLUDE A 15 SECOND DELAY PRIOR TO SWITCHING IN ORDER TO COORDINATE WITH UPSTREAM 15KV PEPPER SWITCHGEAR, KNOFF SWITCH, AND F-SUBSTATION SWITCHGEAR TO ALLOW 15KV SYSTEM TO OPERATE FIRST. UPON 15KV SYSTEM OPERATING, BOTH FEEDERS AT PEPPER 480V GEAR SHOULD BECOME RE-ENERGIZED, AND THEREFORE MAY NOT NEED TO OPERATE.

PROTECTIVE INTERLOCKING:
PARALLELING OF BOTH SOURCES 'A' AND 'B' IS PROHIBITED.

ALL SWITCHING WITHIN 480V GEAR SHALL BE OPEN TRANSITION.

THE FOLLOWING PROTECTIVE INTERLOCKS ARE PROVIDED IN THE MANUAL MODE:
PARALLELING OF SOURCES NOTED ABOVE IS PREVENTED IN THE MANUAL MODE VIA ELECTRICAL INTERLOCKING.

THE FOLLOWING PROTECTIVE INTERLOCKS ARE PROVIDED IN THE AUTOMATIC MODE:
PARALLELING OF THE SOURCES NOTED ABOVE IS NOT PERMITTED.
AUTOMATIC OPERATION IS DISABLED IF:
1. A BREAKER IS RACKED OUT OF THE CONNECTED POSITION.
2. A BREAKER IS TRIPPED OTHER THAN BY THE AUTO-TRANSFER CONTROL.
3. THE AUTO-TRANSFER SWITCH (MANUAL/AUTO) IS SET TO MANUAL MODE.

NORMAL MODE OF OPERATION:
MAIN BREAKER 'A' AND MAIN BREAKER 'B' CLOSED.
TIE BREAKER OPEN.

LOSS OF VOLTAGE ON MAIN 'A' ONLY:
MAIN 'A' UNDERVOLTAGE DEVICE 27 DROPS OUT.
AFTER PROGRAMMABLE TIME DELAY NOTED ABOVE (SET TO 15 SECONDS), MAIN BREAKER 'A' TRIPS OPEN THROUGH AUTOMATIC TRIP CIRCUIT.
AFTER MAIN BREAKER 'A' OPENS AND AFTER A PROGRAMMABLE TIME DELAY (SET TO 2 SECONDS), TIE BREAKER TO CLOSE THROUGH AUTOMATIC CLOSE CIRCUIT.
MAIN 'A' SOURCE IS MONITORED FOR POSITIVE SEQUENCE VOLTAGE.

RESTORATION OF VOLTAGE ON MAIN 'A':
MAIN 'A' UNDERVOLTAGE DEVICE 27 PICKS UP.
FOLLOWING PROGRAMMABLE SOURCE-STABILIZATION TIME PERIOD (SET TO 10 MINUTES), TIE BREAKER TRIPS OPEN THROUGH AUTOMATIC TRIP CIRCUIT.
AFTER TIE BREAKER OPENS, MAIN BREAKER 'A' CLOSSES THROUGH THE AUTOMATIC CLOSE CIRCUIT.

LOSS OF VOLTAGE ON MAIN 'B' ONLY:
MAIN 'B' UNDERVOLTAGE DEVICE 27 DROPS OUT.
AFTER PROGRAMMABLE TIME DELAY NOTED ABOVE (SET TO 15 SECONDS), MAIN BREAKER 'B' TRIPS OPEN THROUGH AUTOMATIC TRIP CIRCUIT.
AFTER MAIN BREAKER 'B' OPENS AND AFTER A PROGRAMMABLE TIME DELAY (SET TO 2 SECONDS), TIE BREAKER TO CLOSE THROUGH AUTOMATIC CLOSE CIRCUIT.
MAIN 'B' SOURCE IS MONITORED FOR POSITIVE SEQUENCE VOLTAGE.

RESTORATION OF VOLTAGE ON MAIN 'B':
MAIN 'B' UNDERVOLTAGE DEVICE 27 PICKS UP.
FOLLOWING PROGRAMMABLE SOURCE-STABILIZATION TIME PERIOD (SET TO 10 MINUTES), TIE BREAKER TRIPS OPEN THROUGH AUTOMATIC TRIP CIRCUIT.
AFTER TIE BREAKER OPENS, MAIN BREAKER 'B' CLOSSES THROUGH THE AUTOMATIC CLOSE CIRCUIT.

SIMULTANEOUS LOSS OF VOLTAGE ON MAIN 'A' AND MAIN 'B':
MAIN 'A' AND 'B' UNDERVOLTAGE DEVICES 27 DROP OUT.
SOURCES 'A' AND 'B' ARE MONITORED FOR POSITIVE SEQUENCE VOLTAGE.
MAIN BREAKERS REMAIN CLOSED AND TIE BREAKER REMAINS OPEN.

SIMULTANEOUS RESTORATION OF VOLTAGE ON MAIN 'A' AND MAIN 'B':
MAIN 'A' AND 'B' UNDERVOLTAGE DEVICES 27 PICK UP.
BOTH MAIN BREAKERS 'A' AND 'B' REMAIN IN CLOSED POSITION WITH TIE BREAKER OPEN.

RESTORATION OF VOLTAGE ON A SINGLE MAIN AFTER LOSS OF BOTH SOURCES ('A'-OR- 'B'):
ASSOCIATED MAIN UNDERVOLTAGE DEVICE 27 PICKS UP.
ASSOCIATED MAIN BREAKER REMAINS CLOSED.
AFTER PROGRAMMABLE TIME DELAY NOTED ABOVE (SET TO 15 SECONDS), OTHER MAIN BREAKER TRIPS OPEN THROUGH AUTOMATIC TRIP CIRCUIT.
AFTER OTHER MAIN BREAKER OPENS AND AFTER A PROGRAMMABLE TIME DELAY (SET TO 2 SECONDS), TIE BREAKER TO CLOSE THROUGH AUTOMATIC CLOSE CIRCUIT.
THE OTHER MAIN SOURCE IS MONITORED FOR POSITIVE SEQUENCE VOLTAGE.

RESTORATION OF VOLTAGE ON REMAINING MAIN:
ASSOCIATED MAIN UNDERVOLTAGE DEVICE 27 PICKS UP.
FOLLOWING PROGRAMMABLE SOURCE-STABILIZATION TIME PERIOD (SET TO 10 MINUTES), TIE BREAKER OPENS THROUGH AUTOMATIC TRIP CIRCUIT.
AFTER TIE BREAKER OPENS, REMAINING UTILITY MAIN BREAKER CLOSSES THROUGH AUTOMATIC CLOSE CIRCUIT.

SCHEDULED OUTAGE NOTES

1. THE CONTRACTOR IS EXPECTED TO PLAN OUT THE EXECUTION OF THE WORK SO AS TO MINIMIZE INTERFERENCE TO THE OWNER'S OPERATIONS. PHASING AND PREPARATION FOR THE WORK SHALL TAKE PLACE SO AS TO ACHIEVE THIS REQUIREMENT.
2. ELECTRICAL POWER OUTAGES TO THE BUILDING IS A REQUIREMENT OF THIS PROJECT AND SHALL BE THOROUGHLY PLANNED AND COORDINATED WITH DMS.
3. PROVIDE COMPLETE PROPOSED CONSTRUCTION SCHEDULE TO ENGINEER AND DMS PROJECT MANAGER FOR REVIEW AND APPROVAL PRIOR TO BEGINNING CONSTRUCTION.
4. ELECTRICAL OUTAGES WILL BE LIMITED TO AFTER HOURS ON NIGHTS AND WEEKENDS. THE CONTRACTOR SHALL PLAN AND STAFF THE PROJECT ACCORDINGLY TO MEET THAT TIMEFRAME.
5. SCHEDULE A MEETING WITH THE ENGINEER, THE OWNER'S PROJECT MANAGER, AND THE CONTRACTOR'S SUPERVISOR WITHIN TWO WEEKS AFTER ACTIVATION TO DISCUSS CONTRACTOR'S STRATEGY FOR OUTAGES AND ACCOMPLISHING THE WORK.
6. COORDINATE ALL POWER OUTAGES WITH DMS PROJECT MANAGER PRIOR TO OUTAGE. PROVIDE 21 DAY MINIMUM NOTICE TO DMS PROJECT MANAGER AS TO WHEN PROPOSED POWER OUTAGES WILL OCCUR TO RECEIVE APPROVAL. ALL POWER OUTAGES AND CHANGEOVERS SHALL BE SCHEDULED WITH THE DMS PROJECT MANAGER AND SCHEDULES STRICTLY ADHERED TO. IN GENERAL, ALL OUTAGES SHALL BE DURING NIGHTS OR WEEKENDS ONLY. OUTAGES DURING NORMAL BUSINESS HOURS WILL NOT BE PERMITTED. OUTAGES THAT ARE NOT SCHEDULED WILL NOT BE ALLOWED.
7. CONTRACTOR SHALL BE RESPONSIBLE FOR TEMPORARY LOW VOLTAGE CONSTRUCTION POWER AND LIGHTING DURING OUTAGES.
8. THE CONTRACTOR SHALL PROVIDE A TRAINED AND KNOWLEDGEABLE STAFF MEMBER REPRESENTATIVE TO CONSULT WITH THE OWNER PRIOR TO ANY SWITCHING, AND TO PERFORM SWITCHING, TEST CIRCUITS, AND DEEM GROUNDED AND SAFE PRIOR TO WORK.
9. COORDINATION WITH DMS MUST OCCUR FOR ENERGIZING AND DE-ENERGIZING FEEDERS.
10. CONTRACTOR IS RESPONSIBLE FOR ALL SWITCHING OF OWNER'S EQUIPMENT AS NEEDED TO COMPLETE SCOPE OF WORK.
11. MULTIPLE SCHEDULED POWER OUTAGES WILL REMOVE NORMAL POWER TO THE BUILDING. BUILDING WILL BE PROVIDED WITH EMERGENCY POWER FROM EXISTING GENERATOR FOR THE DURATION OF THE OUTAGE, WITH EXCEPTION BEING IF ANY WORK TAKES PLACE WITHIN LIFE SAFETY ENERGIZED EQUIPMENT, AT WHICH CASE THE LIFE SAFETY SYSTEM MAY BE DE-ENERGIZED. REFER TO SEQUENCING PLAN FOR QUANTITY AND ESTIMATED DURATION OF OUTAGES.
12. CONTRACTOR SHALL MAINTAIN FUEL DURING ALL OUTAGES TO KEEP BUILDING LIFE SAFETY GENERATOR RUNNING FOR DURATION OF OUTAGES.
13. CONTRACTOR SHALL PROVIDE GENERATOR FUEL FOLLOWING THE OUTAGE TO TOP OFF THE FUEL TANK FOR THE BUILDING GENERATOR.
14. THE BUILDING GENERATOR AND FUEL TANK SIZE IS AS FOLLOWS:
 - PEPPER LIFE SAFETY GENERATOR #1 - 500KW, 500 GALLON TANK
16. CONTRACTOR SHALL INCLUDE IN THE BID AN ALLOWANCE OF \$5,000 FOR FUEL CHARGES FOR GENERATOR NOTED ABOVE AND SHALL COORDINATE TOP OFF OF FUEL STORAGE TANK NOTED ABOVE WITH #2 FUEL OIL AT COMPLETION OF EACH OUTAGE. PORTION OF ALLOWANCE NOT USED SHALL BE RETURNED TO THE OWNER VIA DEDUCTIVE CHANGE ORDER AT COMPLETION OF PROJECT.

480V BREAKER COORDINATION STUDY

MAIN A			
FRAME RATING:	2500A	300KVA TRANSFORMER	
LT PICKUP:	1 (2500A)	FRAME RATING:	800A
LT DELAY:	8	LT PICKUP:	1 (800A)
ST PICKUP:	5 (12500A)	LT DELAY:	2
ST DELAY:	0.1	ST PICKUP:	6 (4800A)
INST:	6 (15000A)	ST DELAY:	0.2
GF PICKUP:	D (800A)	INST:	8 (6400A)
GF DELAY:	0.3		
ERMS:	2 (5000A)	BUSDUCT A/C2	
		FRAME RATING:	2000A
MAIN B		LT PICKUP:	1 (2000A)
FRAME RATING:	2500A	LT DELAY:	4
LT PICKUP:	1 (2500A)	ST PICKUP:	2.5 (5000A)
LT DELAY:	8	ST DELAY:	0.1
ST PICKUP:	5 (12500A)	INST:	4 (8000A)
ST DELAY:	0.1	GF PICKUP:	A (500A)
INST:	6 (15000A)	GF DELAY:	0.1
GF PICKUP:	D (800A)	ERMS:	2 (4000A)
GF DELAY:	0.3		
ERMS:	2 (5000A)	BUSDUCT L/R2	
		FRAME RATING:	2000A
TIE		LT PICKUP:	1 (2000A)
FRAME RATING:	2500A	LT DELAY:	4
LT PICKUP:	1 (2500A)	ST PICKUP:	2.5 (5000A)
LT DELAY:	8	ST DELAY:	0.1
ST PICKUP:	5 (12500A)	INST:	4 (8000A)
ST DELAY:	0.1	GF PICKUP:	A (500A)
INST:	6 (15000A)	GF DELAY:	0.1
GF PICKUP:	D (800A)	ERMS:	2 (4000A)
GF DELAY:	0.3		
ERMS:	2 (5000A)	PANEL BAC1	
		FRAME RATING:	400A
BUSDUCT A/C1		LT PICKUP:	1 (400A)
FRAME RATING:	2000A	LT DELAY:	8
LT PICKUP:	1 (2000A)	ST PICKUP:	5 (2000A)
LT DELAY:	4	ST DELAY:	0.1
ST PICKUP:	2.5 (5000A)	INST:	8 (3200A)
ST DELAY:	0.1		
INST:	4 (8000A)	ATS	
GF PICKUP:	A (500A)	FRAME RATING:	800A
GF DELAY:	0.1	LT PICKUP:	1 (800A)
ERMS:	2 (4000A)	LT DELAY:	8
		ST PICKUP:	4 (3200A)
BUSDUCT L/R1		ST DELAY:	0.1
FRAME RATING:	2000A	INST:	6 (4800A)
LT PICKUP:	1 (2000A)		
LT DELAY:	4		
ST PICKUP:	2.5 (5000A)		
ST DELAY:	0.1		
INST:	4 (8000A)		
GF PICKUP:	A (500A)		
GF DELAY:	0.1		
ERMS:	2 (4000A)		

CODES AND REQUIREMENTS

THE INSTALLATION SHALL COMPLY WITH THE INDICATED EDITION OF THE FOLLOWING CODES AND ORDINANCES. WHERE SPECIFIC EDITION IS NOT INDICATED, COMPLY WITH THE LATEST PUBLISHED EDITION.

NATIONAL FIRE PROTECTION ASSOCIATION - NFPA
NFPA 70 - 2020; NATIONAL ELECTRICAL CODE

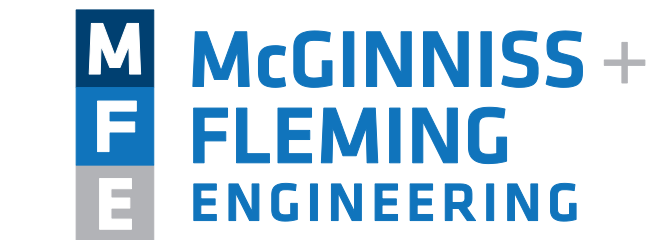
FLORIDA BUILDING CODE
FBC-B 2023; THE FLORIDA BUILDING CODE (8th EDITION)
FPC 2023; THE FLORIDA FIRE PREVENTION CODE (8th EDITION)
INCLUDING NFPA 101 - 2021; THE LIFE SAFETY CODE
FBC-M 2023; THE FLORIDA MECHANICAL CODE (8th EDITION)
FBC-A 2023; THE FLORIDA BUILDING CODE, ACCESSIBILITY (8th EDITION)
FBC-EC 2023; THE FLORIDA BUILDING CODE, ENERGY CONSERVATION (8th EDITION)
FBC-EB 2023; THE FLORIDA BUILDING CODE, EXISTING BUILDING (8th EDITION)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION

IEEE C2; NATIONAL ELECTRICAL SAFETY CODE

SERVING UTILITY COMPANY POLICIES

STATE AND MUNICIPAL CODES AND REQUIREMENTS



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820 EAST PARK AVE, I-200, TALLAHASSEE, FL 32301
MFE-INC.COM | 850.681.6424

PEPPER BUILDING
DESIGN REPLACEMENT OF
MAJOR ELECTRICAL SYSTEMS
PHASE 2

MSFM-02203280

Tallahassee, Florida

DATE:

April 30, 2024

REVISED:

DESIGNED BY: BW	DRAWN BY: TB
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SUBMITTAL:
100% Construction Documents

SHEET TITLE:

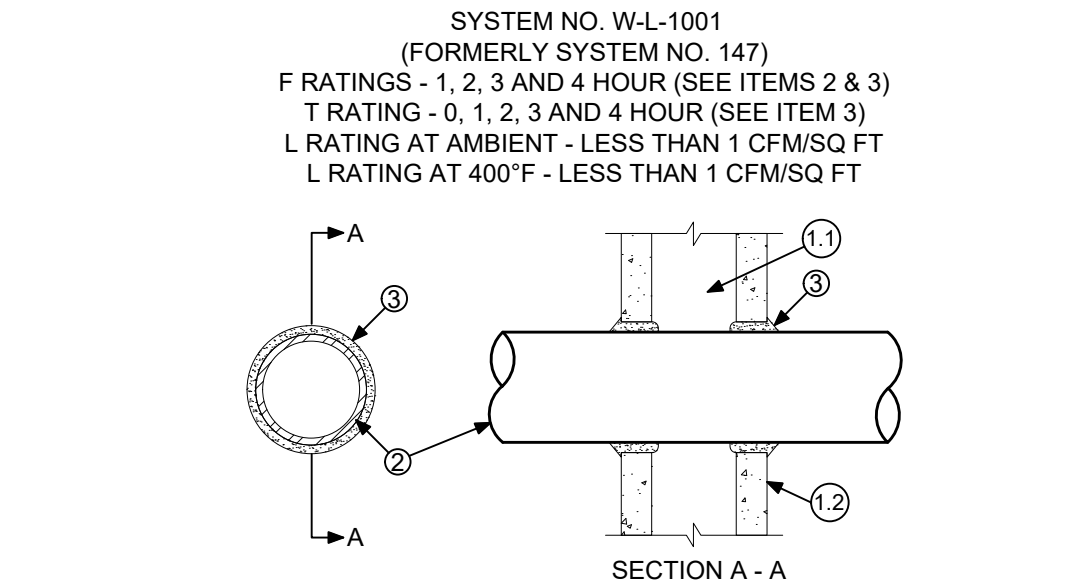
GENERAL NOTES

SHEET:

E101

JOB NUMBER:

2311

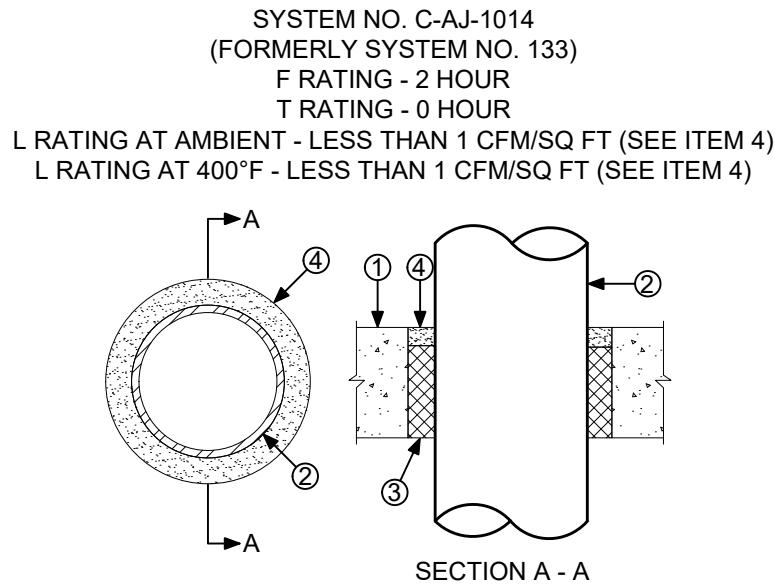


- WALL ASSEMBLY - THE 1, 2, 3 OR 4 HOUR FIRE-RATED GYPSUM WALLBOARD / STUD WALL ASSEMBLY SHALL BE CONSTRUCTED OF THE MATERIALS AND IN THE MANNER DESCRIBED IN THE INDIVIDUAL U300 OR U400 SERIES WALL OR PARTITION DESIGNS IN THE UL FIRE RESISTANCE DIRECTORY AND SHALL INCLUDE THE FOLLOWING CONSTRUCTION FEATURES:
 - STUDS - WALL FRAMING MAY CONSIST OF EITHER WOOD STUDS (MAXIMUM 2 HOUR FIRE RATED ASSEMBLIES) OR STEEL CHANNEL STUDS. WOOD STUDS TO CONSIST OF NOMINAL 2"x4" LUMBER SPACED 16" O.C. WITH NOMINAL 2"x4" LUMBER END PLATES AND CROSS BRACES. STEEL STUDS TO BE MINIMUM 3-5/8" WIDE BY 1-3/8" DEEP CHANNELS SPACED MAXIMUM OF 24" O.C.
 - WALLBOARD, GYPSUM - NOMINAL 1/2" OR 5/8" THICK, 4" WIDE WITH SQUARE OR TAPERED EDGES. THE GYPSUM WALLBOARD TYPE, THICKNESS, NUMBER OF LAYERS, FASTENER TYPE AND SHEET ORIENTATION SHALL BE AS SPECIFIED IN THE INDIVIDUAL U300 OR U400 SERIES DESIGN IN THE UL FIRE RESISTANCE DIRECTORY. MAXIMUM DIAMETER OF OPENING IS 13-1/2".
- PIPE OR CONDUIT - NOMINAL 12" DIAMETER (OR SMALLER) SCHEDULE 10 (OR HEAVIER) STEEL PIPE, NOMINAL 12" DIAMETER (OR SMALLER) SERVICE WEIGHT (OR HEAVIER) CAST IRON SOIL PIPE, NOMINAL 12" DIAMETER (OR SMALLER) CLASS 50 (OR HEAVIER) DUCTILE IRON PRESSURE PIPE, NOMINAL 8" DIAMETER (OR SMALLER) STEEL CONDUIT, NOMINAL 4" DIAMETER (OR SMALLER) STEEL ELECTRICAL METALLIC TUBING, NOMINAL 6" DIAMETER (OR SMALLER) TYPE L (OR HEAVIER) COPPER TUBING OR NOMINAL 1" DIAMETER (OR SMALLER) FLEXIBLE STEEL CONDUIT. WHEN COPPER PIPE IS USED, MAXIMUM F RATING OF FIRESTOP SYSTEM (ITEM 3) IS 2 HOUR. STEEL PIPES OR CONDUITS LARGER THAN NOMINAL 4" DIAMETER MAY ONLY BE USED IN WALLS CONSTRUCTED USING STEEL CHANNEL STUDS. A MAXIMUM OF ONE PIPE OR CONDUIT IS PERMITTED IN THE FIRESTOP SYSTEM. PIPE OR CONDUIT TO BE INSTALLED NEAR CENTER OF STUD CAVITY WIDTH AND TO BE RIGIDLY SUPPORTED ON BOTH SIDES OF WALL ASSEMBLY.
- FILL, VOID OR CAVITY MATERIAL - CAULK - CAULK FILL MATERIAL INSTALLED TO COMPLETELY FILL ANNULAR SPACE BETWEEN PIPE OR CONDUIT AND GYPSUM WALLBOARD AND WITH A MINIMUM 1/4" DIAMETER BEAD OF CAULK APPLIED TO PERIMETER OF PIPE OR CONDUIT AT ITS EGRESS FROM THE WALL. CAULK INSTALLED SYMMETRICALLY ON BOTH SIDES OF WALL ASSEMBLY. THE HOURLY F RATING OF THE FIRESTOP SYSTEM IS DEPENDENT UPON THE HOURLY FIRE RATING OF THE WALL ASSEMBLY IN WHICH IT IS INSTALLED, AS SHOWN IN THE FOLLOWING TABLE. THE HOURLY T RATING OF THE FIRESTOP SYSTEM IS DEPENDENT UPON THE TYPE OR SIZE OF PIPE OR CONDUIT AND THE HOURLY FIRE RATING OF THE WALL ASSEMBLY IN WHICH IT IS INSTALLED, AS TABULATED BELOW:

MAXIMUM PIPE DIAMETER (INCHES)	ANNULAR SPACE (INCHES)	F RATING (HOURS)	T RATING (HOURS)
4	0 - 1-1/2	1 OR 2	0

PENETRATION DETAIL - A

NOT TO SCALE

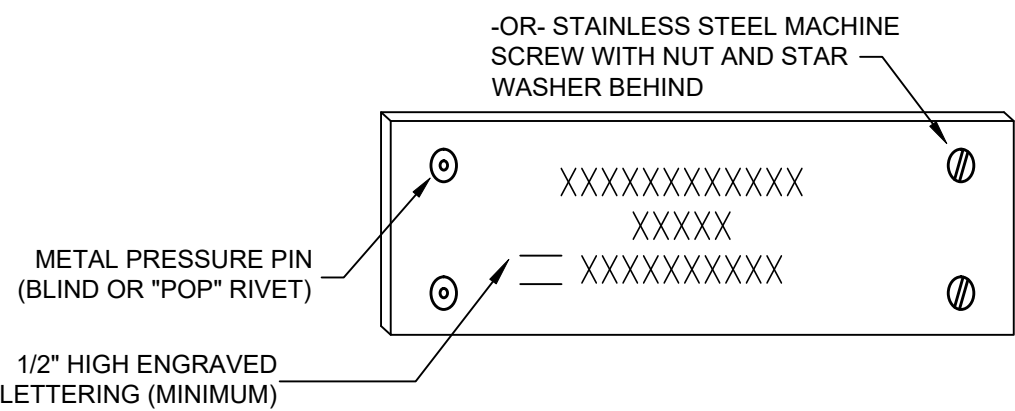


- FLOOR OR WALL ASSEMBLY - MIN 3-1/4" THICK REINFORCED LIGHTWEIGHT OR NORMAL WEIGHT (100-150 PCF) CONCRETE. WALL MAY ALSO BE CONSTRUCTED OF ANY UL CLASSIFIED CONCRETE BLOCKS*. MAX DIAMETER OF CIRCULAR OPENING IS 6". SEE CONCRETE BLOCK (CAZT) CATEGORY IN THE FIRE RESISTANCE DIRECTORY FOR NAMES OF MANUFACTURERS.
- THROUGH PENETRANTS - ONE METALLIC PIPE OR CONDUIT TO BE CENTERED WITHIN THE FIRESTOP SYSTEM. A NOMINAL ANNULAR SPACE OF 3/4" IS REQUIRED WITHIN THE FIRESTOP SYSTEM. PIPE OR CONDUIT TO BE RIGIDLY SUPPORTED ON BOTH SIDES OF FLOOR OR WALL ASSEMBLY. THE FOLLOWING TYPES AND SIZES OF METALLIC PIPES OR CONDUITS MAY BE USED:
 - STEEL PIPE - NOMINAL 4" DIAMETER (OR SMALLER) SCHEDULE 10 (OR HEAVIER) STEEL PIPE.
 - CONDUIT - NOMINAL 4" DIAMETER (OR SMALLER) STEEL ELECTRICAL METALLIC TUBING OR STEEL CONDUIT.
- PACKING MATERIAL - MINIMUM 4 PCF MINERAL WOOL BATT INSULATION FIRMLY PACKED INTO OPENING AS A PERMANENT FORM. PACKING MATERIAL TO BE RECESSED FROM TOP SURFACE OF FLOOR OR FROM BOTH SURFACES OF WALL AS REQUIRED TO ACCOMMODATE THE REQUIRED THICKNESS OF FILL MATERIAL. MINIMUM THICKNESS OF PACKING MATERIAL IN FLOORS AND WALLS TO BE 2-3/4" AND 2-1/4", RESPECTIVELY.
- FILL, VOID OR CAVITY MATERIAL* - SEALANT - MINIMUM 1/2" THICKNESS OF FILL MATERIAL APPLIED WITHIN ANNULUS, FLUSH WITH TOP SURFACE OF FLOOR OR WITH BOTH SURFACES OF WALL. AS AN ALTERNATE, THE PERMANENT FORMING MATERIAL (ITEM 3) MAY BE OMITTED IF THE FILL MATERIAL THICKNESS IS INCREASED TO A MINIMUM OF 1-1/2".
MINNESOTA MINING AND MFG CO. - (TYPES FB-2000 OR FB-2000+). (NOTE: L RATINGS APPLY ONLY WHEN FB-2000+ IS USED.)

*BEARING THE UL CLASSIFICATION MARKING.

PENETRATION DETAIL - B

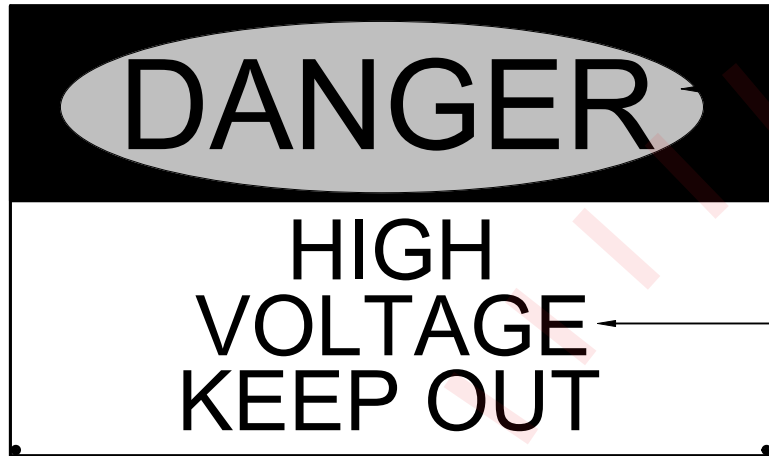
NOT TO SCALE



EQUIPMENT IDENTIFICATION SHALL BE MADE USING ENGRAVED LAMINATED PLASTIC PLATES (INDENTED TAPE LABELS WILL NOT BE PERMITTED). CHARACTERS SHALL BE WHITE ON A BLACK BACKGROUND AND 1/2" HIGH MINIMUM. PLATES SHALL BE SECURED TO THE PANELS BY MEANS OF SCREWS OR METAL PRESSURE PINS. ADHESIVES FASTENING ALONE WILL NOT BE ACCEPTABLE. ALL NAMEPLATES SHALL BE MOUNTED ON THE OUTSIDE SURFACE OF THE PIECE OF EQUIPMENT.

NAMEPLATE DETAIL

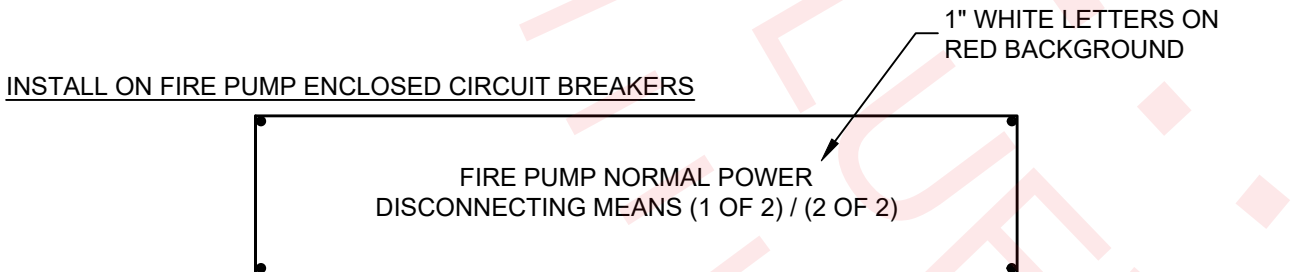
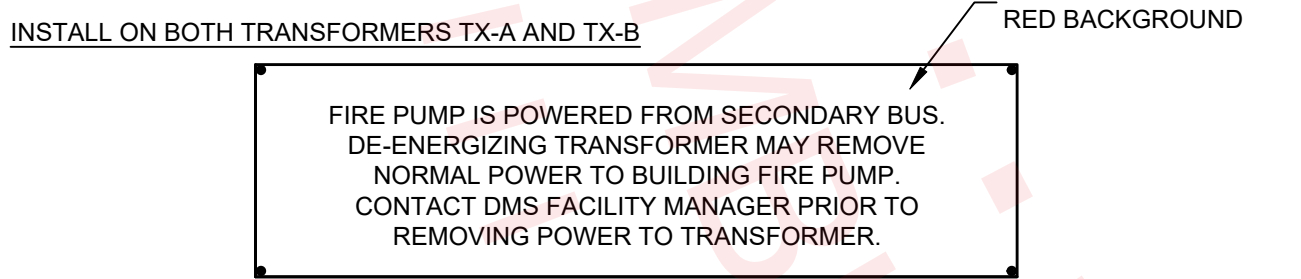
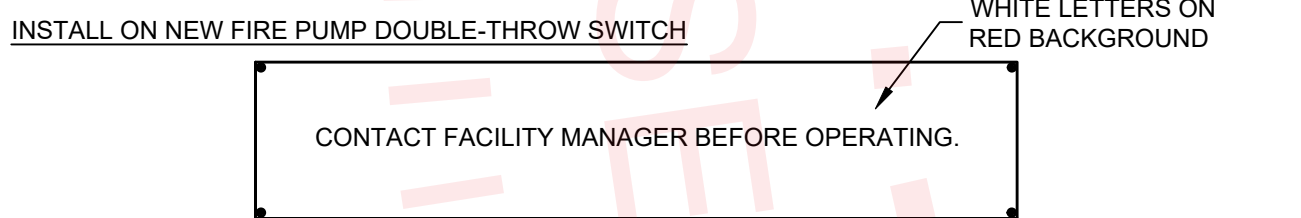
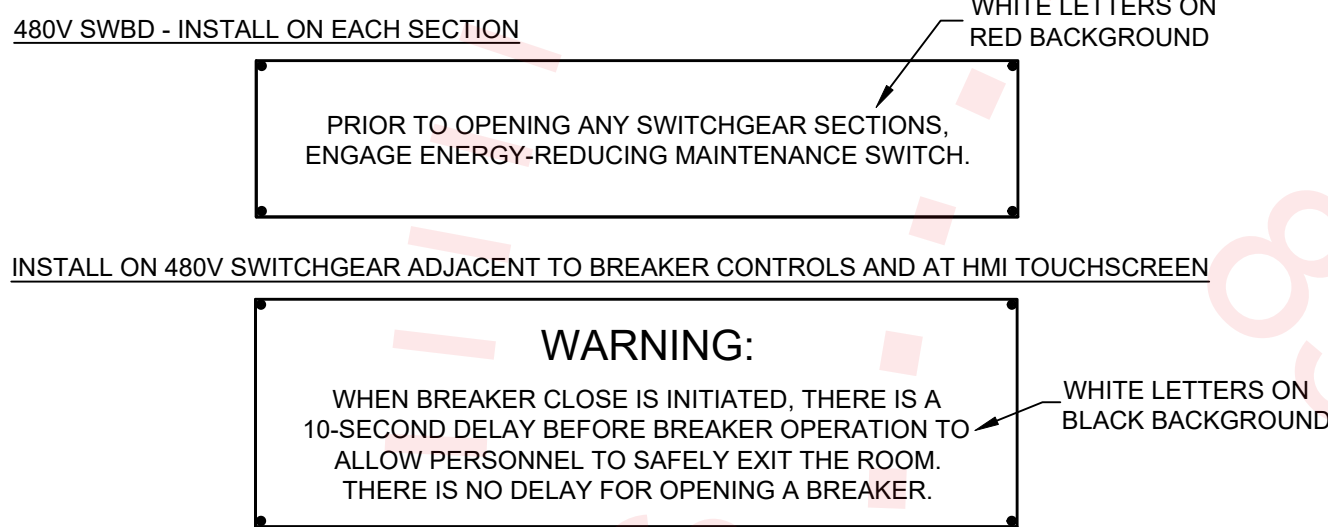
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PROVIDE NEW WARNING LABELS FOR EACH DOOR OF SWITCHGEAR

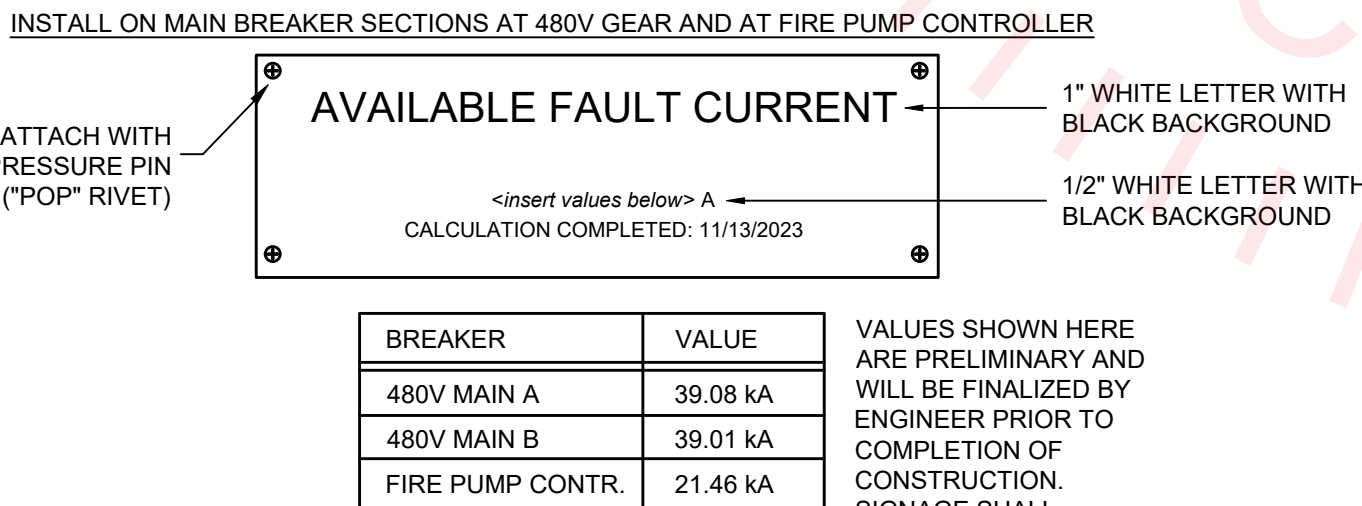
WARNING PLATE DETAIL

NOT TO SCALE



NAMEPLATE DETAILS

NOT TO SCALE



VALUES SHOWN HERE ARE PRELIMINARY AND WILL BE FINALIZED BY ENGINEER PRIOR TO COMPLETION OF CONSTRUCTION. SIGNAGE SHALL INCLUDE FINAL VALUES.

FAULT CURRENT NAMEPLATE DETAIL

NOT TO SCALE

ARC FLASH STUDY CLARIFICATION

NOTE: Arc flash labels were created using SKM Powertools® as the arc flash analysis program. Overcurrent protective device modeling files for specific device manufacturer and model were provided by SKM library.

The equations used for the calculation of the estimated maximum incident energy and the estimated arc flash boundary distance are based upon measured incident energy under a specific set of test conditions and on theoretical work. Actual arc flash exposures may be more or less severe than calculated by the arc flash analysis program. Calculations are based on the installation and configuration of the electrical distribution system as initially constructed. Any changes to the electrical distribution system may cause these calculations to be invalid. All short circuit and overcurrent protective devices (OCPD) are assumed to operate as originally intended by manufacturer.

It should be noted that the personal protective equipment (PPE) for the arc flash hazard is the last line-of-defense. It is not intended nor will it prevent all injuries. NFPA 70E PPE levels are intended to reduce the impact of an arc flash to 2nd degree burns for the torso and head only. NFPA 70E states that the incident energy exposure shall be based on the working distance of the employee's face and head. Objects closer to the arc flash will be exposed to much greater levels of incident energy. Fire rated (FR) clothing and PPE shall be used based upon the incident energy exposure. This means injuries to hands and arms may be expected if an arc at the level calculated and protected against occurs.

Incident energy levels are directly related to the clearing time of the upstream OCPD. Operation of the OCPD within the manufacturers design specifications is essential for limiting the incident energy due to arc flash hazards. Failure of the OCPD components to operate within the manufacturer's time current curves will compromise the results of this arc flash study and will result in higher levels of incident energy to electrical workers. NFPA 70E standards recommend regularly scheduled maintenance and testing be performed on the electrical distribution system components to assure proper operation of all overcurrent protective devices. Regular testing is required to protect electrical workers from greater arc flash energy hazards than are calculated in this study.

WARNING

Arc Flash and Shock Risks Appropriate PPE Required

ARC FLASH RISK PROTECTION	PPE
Incident Energy: 15.4 cal/cm² Arc Flash Boundary: 89 in 480 VAC Shock Risk When Cover is Removed	• 25 cal/sq cm, Arc-rated shirt (long-sleeve) plus Arc-rated pants (long) plus Arc-rated coverall, plus arc rated arc flash suit jacket, pants, & hood, Arc-rated rainwear as needed. • Hardhat + Arc-rated hard hat liner + Safety Glasses or Goggles + Ear Canal Inserts • Arc-rated Gloves • Leather work shoes
Shock Protection Boundaries: Limited Approach: 42 in Restricted Approach: 12 in	
Equipment ID: 480V BUS A	Date of Analysis: 04/25/2024

WARNING

Arc Flash and Shock Risks Appropriate PPE Required

ARC FLASH RISK PROTECTION	PPE
Incident Energy: 15.4 cal/cm² Arc Flash Boundary: 89 in 480 VAC Shock Risk When Cover is Removed	• 25 cal/sq cm, Arc-rated shirt (long-sleeve) plus Arc-rated pants (long) plus Arc-rated coverall, plus arc rated arc flash suit jacket, pants, & hood, Arc-rated rainwear as needed. • Hardhat + Arc-rated hard hat liner + Safety Glasses or Goggles + Ear Canal Inserts • Arc-rated Gloves • Leather work shoes
Shock Protection Boundaries: Limited Approach: 42 in Restricted Approach: 12 in	
Equipment ID: 480V BUS B	Date of Analysis: 04/25/2024

WARNING

Arc Flash and Shock Risks Appropriate PPE Required

ARC FLASH RISK PROTECTION	PPE
Incident Energy: 12.7 cal/cm² Arc Flash Boundary: 79 in 480 VAC Shock Risk When Cover is Removed	• 25 cal/sq cm, Arc-rated shirt (long-sleeve) plus Arc-rated pants (long) plus Arc-rated coverall, plus arc rated arc flash suit jacket, pants, & hood, Arc-rated rainwear as needed. • Hardhat + Arc-rated hard hat liner + Safety Glasses or Goggles + Ear Canal Inserts • Arc-rated Gloves • Leather work shoes
Shock Protection Boundaries: Limited Approach: 42 in Restricted Approach: 12 in	
Equipment ID: FIRE PUMP CTRL.	Date of Analysis: 04/25/2024

WARNING

Arc Flash and Shock Risks Appropriate PPE Required

ARC FLASH RISK PROTECTION	PPE
Incident Energy: 0.25 cal/cm² Arc Flash Boundary: 7 in 480 VAC Shock Risk When Cover is Removed	• Non-melting or untreated natural fiber (cotton/wool/rayon/silk > 4.5 oz/sq yd), shirt (long-sleeve), pants (long). • Safety Glasses or Goggles + Ear Canal Inserts • Leather Gloves • N/A
Shock Protection Boundaries: Limited Approach: 42 in Restricted Approach: 12 in	
Equipment ID: 1AC1	Date of Analysis: 04/25/2024

WARNING

Arc Flash and Shock Risks Appropriate PPE Required

ARC FLASH RISK PROTECTION	PPE
Incident Energy: 0.22 cal/cm² Arc Flash Boundary: 6 in 480 VAC Shock Risk When Cover is Removed	• Non-melting or untreated natural fiber (cotton/wool/rayon/silk > 4.5 oz/sq yd), shirt (long-sleeve), pants (long). • Safety Glasses or Goggles + Ear Canal Inserts • Leather Gloves • N/A
Shock Protection Boundaries: Limited Approach: 42 in Restricted Approach: 12 in	
Equipment ID: 2AC1	Date of Analysis: 04/25/2024

WARNING

Arc Flash and Shock Risks Appropriate PPE Required

ARC FLASH RISK PROTECTION	PPE
Incident Energy: 0.22 cal/cm² Arc Flash Boundary: 6 in 480 VAC Shock Risk When Cover is Removed	• Non-melting or untreated natural fiber (cotton/wool/rayon/silk > 4.5 oz/sq yd), shirt (long-sleeve), pants (long). • Safety Glasses or Goggles + Ear Canal Inserts • Leather Gloves • N/A
Shock Protection Boundaries: Limited Approach: 42 in Restricted Approach: 12 in	
Equipment ID: 3AC1	Date of Analysis: 04/25/2024

WARNING

Arc Flash and Shock Risks Appropriate PPE Required

ARC FLASH RISK PROTECTION	PPE
Incident Energy: 0.22 cal/cm² Arc Flash Boundary: 6 in 480 VAC Shock Risk When Cover is Removed	• Non-melting or untreated natural fiber (cotton/wool/rayon/silk > 4.5 oz/sq yd), shirt (long-sleeve), pants (long). • Safety Glasses or Goggles + Ear Canal Inserts • Leather Gloves • N/A
Shock Protection Boundaries: Limited Approach: 42 in Restricted Approach: 12 in	
Equipment ID: 4AC1	Date of Analysis: 04/25/2024

WARNING

Arc Flash and Shock Risks Appropriate PPE Required

ARC FLASH RISK PROTECTION	PPE
Incident Energy: 0.22 cal/cm² Arc Flash Boundary: 6 in 480 VAC Shock Risk When Cover is Removed	• Non-melting or untreated natural fiber (cotton/wool/rayon/silk > 4.5 oz/sq yd), shirt (long-sleeve), pants (long). • Safety Glasses or Goggles + Ear Canal Inserts • Leather Gloves • N/A
Shock Protection Boundaries: Limited Approach: 42 in Restricted Approach: 12 in	
Equipment ID: 5AC1	Date of Analysis: 04/25/2024

WARNING

Arc Flash and Shock Risks Appropriate PPE Required

ARC FLASH RISK PROTECTION	PPE
Incident Energy: 0.22 cal/cm² Arc Flash Boundary: 6 in 480 VAC Shock Risk When Cover is Removed	• Non-melting or untreated natural fiber (cotton/wool/rayon/silk > 4.5 oz/sq yd), shirt (long-sleeve), pants (long). • Safety Glasses or Goggles + Ear Canal Inserts • Leather Gloves • N/A
Shock Protection Boundaries: Limited Approach: 42 in Restricted Approach: 12 in	
Equipment ID: 6AC1	Date of Analysis: 04/25/2024

WARNING

Arc Flash and Shock Risks Appropriate PPE Required

ARC FLASH RISK PROTECTION	PPE
Incident Energy: 0.22 cal/cm² Arc Flash Boundary: 6 in 480 VAC Shock Risk When Cover is Removed	• Non-melting or untreated natural fiber (cotton/wool/rayon/silk > 4.5 oz/sq yd), shirt (long-sleeve), pants (long). • Safety Glasses or Goggles + Ear Canal Inserts • Leather Gloves • N/A
Shock Protection Boundaries: Limited Approach: 42 in Restricted Approach: 12 in	
Equipment ID: 7AC1	Date of Analysis: 04/25/2024

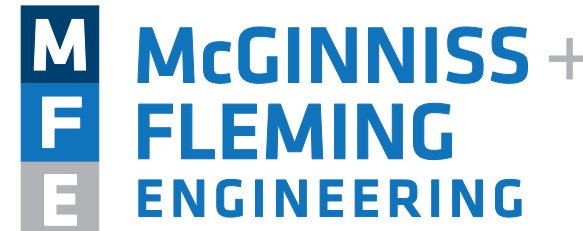
WARNING

Arc Flash and Shock Risks Appropriate PPE Required

ARC FLASH RISK PROTECTION	PPE
Incident Energy: 0.21 cal/cm² Arc Flash Boundary: 6 in 480 VAC Shock Risk When Cover is Removed	• Non-melting or untreated natural fiber (cotton/wool/rayon/silk > 4.5 oz/sq yd), shirt (long-sleeve), pants (long). • Safety Glasses or Goggles + Ear Canal Inserts • Leather Gloves • N/A
Shock Protection Boundaries: Limited Approach: 42 in Restricted Approach: 12 in	
Equipment ID: 8AC1	Date of Analysis: 04/25/2024

ARC FLASH LABELS

NOT TO SCALE



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MFE-INC.COM | 850.681.6424

PEPPER BUILDING
DESIGN REPLACEMENT OF
MAJOR ELECTRICAL SYSTEMS
PHASE 2
MSFM-02203280
Tallahassee, Florida

DATE:
April 30, 2024

REVISED:

DESIGNED BY: BW
DRAWN BY: TB

SUBMITTAL:
100% Construction Documents

SHEET TITLE:
ELECTRICAL DETAILS

SHEET:

E102

JOB NUMBER:
2311

PHASE 1 (INITIAL)	PHASE 2 (OUTAGE #1)	PHASE 3 (OUTAGE #2)
1. CONTRACTOR SHALL VISIT EACH GEAR TO BE OPERATED AS PART OF PROJECT AND CONFIRM INITIAL CONDITIONS ARE AS NOTED. ANY DISCREPANCIES SHALL BE IMMEDIATELY BROUGHT TO THE ENGINEER'S ATTENTION AND SWITCHING PLAN SHALL BE MODIFIED BY CONTRACTOR AS NEEDED. 2. CONTRACTOR SHALL SPEND TIME ONSITE REVIEWING PLANS AND BECOMING FAMILIAR WITH EACH SCOPE OF WORK AND ALL EQUIPMENT AFFECTED BY SCOPE OF WORK. 3. NOTE THAT PANELBOARDS NOTED TO BE REPLACED ARE NOT SPECIFICALLY SEQUENCED IN THE FOLLOWING PHASES. CONTRACTOR SHALL COORDINATE WITH OUTAGE TIMES AVAILABLE TO REPLACE PANELS DURING BUILDING OUTAGES, COMPLETING AS MANY PANEL REPLACEMENTS AS TIME PERMITS DURING ALL SCHEDULED OUTAGES UNTIL ALL PANELS ARE COMPLETED.	<< 10 HOUR OUTAGE FOR PEPPER BUILDING >> 1. OPEN 15KV TX-A AND TX-B FEEDER BREAKERS AT 15KV SWGR. OLITS TO REMAIN ENERGIZED. 2. LIFE SAFETY GENERATOR WILL START TO PICKUP RESPECTIVE LOADS. 3. VERIFY 480V SWGR IS DE-ENERGIZED. 4. 480V SWGR RETROFIL TEAM SHALL PERFORM INVESTIGATION AND MEASUREMENTS EFFORT OF EXISTING SWGR LINEUP FOR USE WITH ORDERING MATERIALS. 5. INSTALL FIRE PUMP A-SIDE ECB, B-SIDE ECB, MANUAL TRANSFER SWITCH, AND NEW FEEDERS BETWEEN TRANSFORMERS AND MANUAL TRANSFER SWITCH. LEAVE BOTH ECB's OPEN AND LOCKED OUT. 6. CLOSE 15KV TX-A AND TX-B FEEDER BREAKERS AT 15KV SWGR. 7. VERIFY LIFE SAFETY ATS TRANSFERS BACK TO NORMAL AND THAT GENERATOR SHUTS DOWN. << POWER IS RESTORED >> 8. COMPLETE REFUELING OF GENERATOR TANK.	<< 48 HOUR OUTAGE FOR PEPPER BUILDING >> 1. OPEN 15KV TX-A AND TX-B FEEDER BREAKERS AT 15KV SWGR. OLITS TO REMAIN ENERGIZED. 2. LIFE SAFETY GENERATOR WILL START TO PICKUP RESPECTIVE LOADS. 3. VERIFY 480V SWGR IS DE-ENERGIZED. 4. BEGIN 480V SWGR RETROFIL SCOPE OF WORK. 5. INSTALL FEEDER IN RACEWAY FROM MANUAL TRANSFER SWITCH TO FIRE PUMP CONTROLLER. COORDINATE WITH CONTROLLER FOR RACEWAY / FEEDER TIE-IN. NOTE THAT EMERGENCY POWER WILL STILL BE ENERGIZED WITHIN FIRE PUMP CONTROLLER. CONTRACTOR TO OPEN EMERGENCY BREAKER SERVING FIRE PUMP TO DE-ENERGIZE EQUIPMENT WHEN WORKING INSIDE CABINET. DEPENDING UPON LENGTH OF TIME TO COMPLETE WORK, FIRE WATCH MAY BE REQUIRED, AND EMERGENCY SOURCE SHALL REMAIN READY AND AVAILABLE TO REFEED FIRE PUMP CONTROLLER IF NEEDED. 6. REMOVE FEEDER BETWEEN 480V SWITCHGEAR AND FIRE PUMP CONTROLLER AND SEAL REMAINING CONDUITS AT FLOOR. 7. UNLOCK AND CLOSE BOTH FIRE PUMP ECB's. 8. INSTALL GROUNDING FROM EXISTING 480V SWITCHGEAR TO GROUND BUS IN ELECTRICAL ROOM. 9. CLOSE 15KV TX-A AND TX-B FEEDER BREAKERS AT 15KV SWGR. 10. VERIFY LIFE SAFETY ATS TRANSFERS BACK TO NORMAL AND THAT GENERATOR SHUTS DOWN. 11. VERIFY FIRE PUMP CONTROLLER HAS UTILITY POWER AVAILABLE. << POWER IS RESTORED >> 12. COMPLETE REFUELING OF GENERATOR TANK.
PHASE 4 (OUTAGE #3)	PHASE 5 (OUTAGE #4)	PHASE 6 (OUTAGE #5)
<< 48 HOUR OUTAGE FOR PEPPER BUILDING >> 1. OPEN 15KV TX-A AND TX-B FEEDER BREAKERS AT 15KV SWGR. OLITS TO REMAIN ENERGIZED. 2. LIFE SAFETY GENERATOR WILL START TO PICKUP RESPECTIVE LOADS. 3. VERIFY 480V SWGR IS DE-ENERGIZED. 4. CONTINUE WITH 480V SWGR RETROFIL SCOPE OF WORK. 5. CLOSE 15KV TX-A AND TX-B FEEDER BREAKERS AT 15KV SWGR. 6. VERIFY LIFE SAFETY ATS TRANSFERS BACK TO NORMAL AND THAT GENERATOR SHUTS DOWN. << POWER IS RESTORED >> 7. COMPLETE REFUELING OF GENERATOR TANK.	<< 48 HOUR OUTAGE FOR PEPPER BUILDING >> 1. OPEN 15KV TX-A AND TX-B FEEDER BREAKERS AT 15KV SWGR. OLITS TO REMAIN ENERGIZED. 2. LIFE SAFETY GENERATOR WILL START TO PICKUP RESPECTIVE LOADS. 3. VERIFY 480V SWGR IS DE-ENERGIZED. 4. CONTINUE WITH 480V SWGR RETROFIL SCOPE OF WORK. 5. CLOSE 15KV TX-A AND TX-B FEEDER BREAKERS AT 15KV SWGR. 6. VERIFY LIFE SAFETY ATS TRANSFERS BACK TO NORMAL AND THAT GENERATOR SHUTS DOWN. << POWER IS RESTORED >> 7. COMPLETE REFUELING OF GENERATOR TANK.	<< 48 HOUR OUTAGE FOR PEPPER BUILDING >> 1. OPEN 15KV TX-A AND TX-B FEEDER BREAKERS AT 15KV SWGR. OLITS TO REMAIN ENERGIZED. 2. LIFE SAFETY GENERATOR WILL START TO PICKUP RESPECTIVE LOADS. 3. VERIFY 480V SWGR IS DE-ENERGIZED. 4. CONTINUE WITH 480V SWGR RETROFIL SCOPE OF WORK. 5. INSTALL 480V SWGR HMI WALL-MOUNT CABINET AND ALL WIRING IN RACEWAY. 6. AFTER COMPLETION OF RETROFIL SCOPE, DURING THIS OUTAGE PERFORM TESTING, DEMONSTRATION, AND OWNER TRAINING OF SWGR CONTROLS. 7. CLOSE 15KV TX-A AND TX-B FEEDER BREAKERS TO RESTORE POWER TO 480V SWGR. VERIFY ALL LOADS ARE ON NORMAL POWER. 8. VERIFY LIFE SAFETY ATS TRANSFERS BACK TO NORMAL AND LIFE SAFETY GENERATOR SHUTS DOWN. << POWER IS RESTORED >> 9. COMPLETE REFUELING OF GENERATOR TANK. 10. COMPLETE ALL SCADA CONNECTIONS AND IMPLEMENT SCADA CHANGES INTO PME. 11. COMPLETE SWGR PANEL PAINTING, ALL SIGNAGE, AND SITE CLEAN-UP.

- SHEET NOTES:
- FLOORPLAN CONFIGURATIONS SHOWN FOR EACH PHASE INDICATE CONDITION FOLLOWING PHASE SCOPE OF WORK.
 - PHASING DIAGRAMS AND NOTES DO NOT INCLUDE ALL DETAILS OF CONSTRUCTION.
 - CONTRACTOR SHALL BE RESPONSIBLE FOR PERFORMING ALL WORK IN A SAFE MANNER AND COMPLETING ALL NECESSARY LOCK-OUT, TAG-OUT PROCEDURES.
 - CONTRACTOR SHALL BE RESPONSIBLE FOR CONFIRMING ALL SEQUENCING, SWITCHING, AND REQUIRED ACTIONS FOR SAFELY PERFORMING ALL WORK NOTED.
 - MULTIPLE TESTING AND START-UP SERVICES WILL BE A REQUIREMENT OF THIS PROJECT TO ALLOW FOR EQUIPMENT REPLACEMENT IN PHASES.
 - SWITCHING OF DMS EQUIPMENT AS LISTED IN SEQUENCING NOTES SHALL BE COMPLETED BY ELECTRICAL CONTRACTOR.
 - CONTRACTOR SHALL COORDINATE SWITCHING SCHEDULE WITH DMS.
 - SEQUENCING NOTES WERE DEVELOPED AT TIME OF DESIGN WITHOUT ANY COORDINATION WITH CONTRACTOR. SHOULD CONTRACTOR HAVE IDEAS TO SIMPLIFY CONSTRUCTION, PROVIDE SAFER WORK ENVIRONMENT, OR TO REDUCE OUTAGE QUANTITIES, CONTRACTOR SHALL PRESENT PLANS TO OWNER AND ENGINEER FOR DISCUSSION AND CONSIDERATION.
 - NOTE THAT SEQUENCING NOTES WERE DEVELOPED WITHOUT KNOWLEDGE OF QUANTITY OF OUTAGES TO BE REQUIRED BY SWITCHGEAR RETROFILL TEAM. SEQUENCING ACCOUNTS FOR AN INITIAL OUTAGE FOR MEASUREMENTS AND INVESTIGATION, FOLLOWED BY (4) WEEKEND OUTAGES TO ACCOMPLISH THE SCOPE OF WORK. ALL OUTAGES SHALL OCCUR ON WEEKENDS, AND MUST BE APPROVED BY THE OWNER.



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PEPPER BUILDING
DESIGN REPLACEMENT OF
MAJOR ELECTRICAL SYSTEMS
PHASE 2
MSFM-02203280
Tallahassee, Florida

DATE:
April 30, 2024

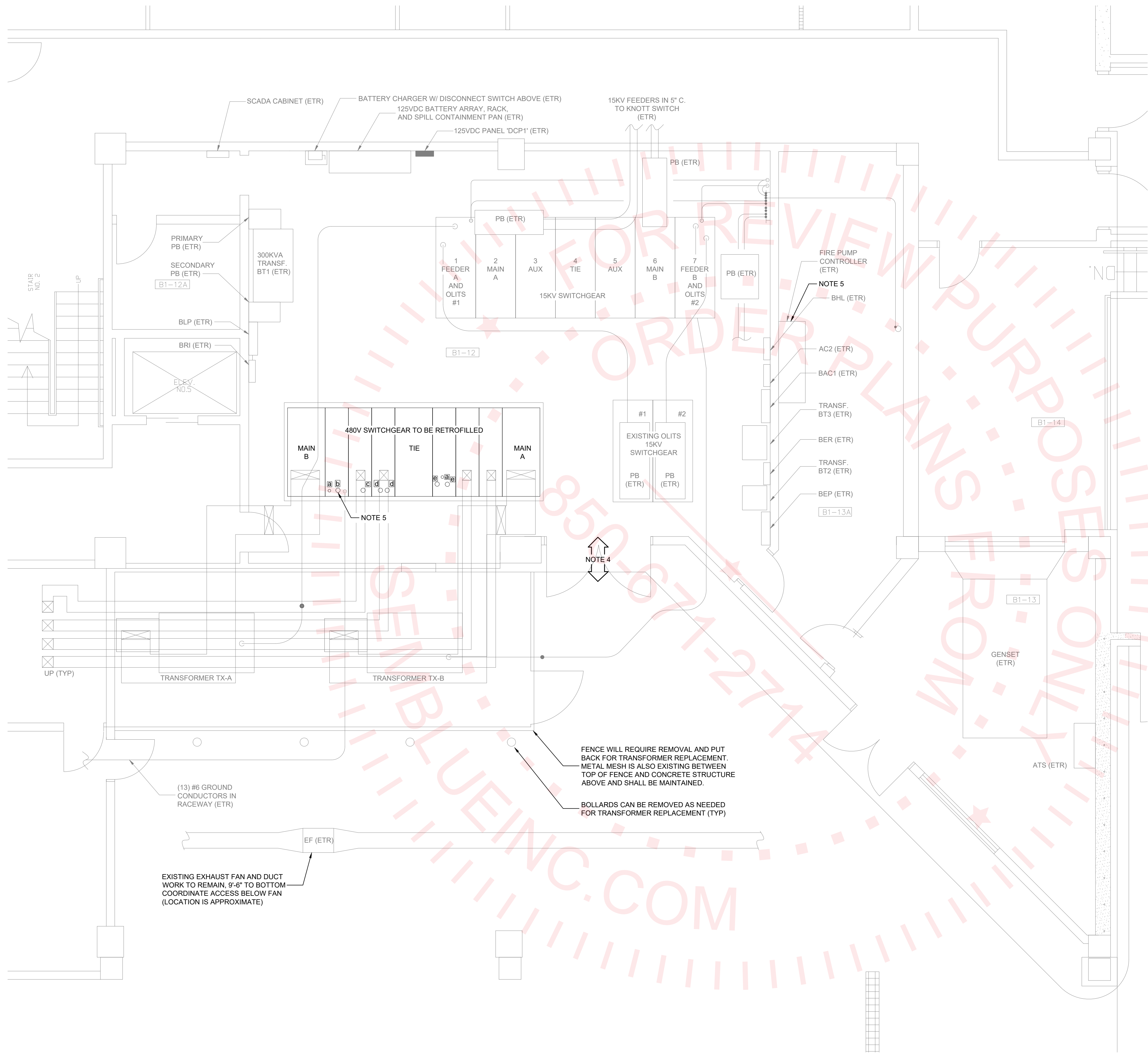
REVISED:

DESIGNED BY: BW	DRAWN BY: TB
--------------------	-----------------

SUBMITTAL:
100% Construction Documents

SHEET TITLE:
SEQUENCING

SHEET:
E103
JOB NUMBER:
2311



SHEET NOTES:

1. REFER TO SEQUENCING NOTES FOR PHASING OF EQUIPMENT REPLACEMENT.
2. NO RACEWAY SHALL BE ABANDONED. ALL RACEWAY THAT IS PART OF EQUIPMENT BEING REMOVED SHALL BE REMOVED IN ITS ENTIRETY, UNLESS NOTED OTHERWISE.
3. NOTE THERE IS MUCH CONDUIT ROUTED OVERHEAD IN ELECTRICAL ROOM THAT IS NOT SERVED BY SWITCHGEAR BEING RETROFILLED AND IS NOT SHOWN HERE, BUT CONTRIBUTES TO CREATE A CONGESTED AREA ABOVE AND AROUND SWITCHGEAR. CONTRACTOR SHALL USE CAUTION WHEN WORKING AROUND FEEDERS / RACEWAY TO REMAIN.
4. EXISTING DOUBLE DOORS SHALL BE USED FOR ACCESS INTO ELECTRICAL ROOM FOR EQUIPMENT REPLACEMENT. ELECTRICAL ROOM FLOOR ELEVATION IS 48\"/>
5. ONCE EXISTING FIRE PUMP NORMAL FEEDER IS REMOVED BETWEEN FIRE PUMP CONTROLLER AND SWITCHGEAR, REMAINING CONDUIT STUB-UP IN FLOOR SHALL BE FILLED WITH GROUT TO CLOSE OFF RACEWAY BETWEEN PUMP CONTROLLER AND GEAR. TYPICAL FOR EACH END.



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**PEPPER BUILDING
DESIGN REPLACEMENT OF
MAJOR ELECTRICAL SYSTEMS
PHASE 2**

MSFM-02203280

Tallahassee, Florida

DATE:
April 30, 2024

REVISED:

DESIGNED BY:
BW

DRAWN BY:
TB

SUBMITTAL:
100% Construction Documents

SHEET TITLE:

DEMOLITION
POWER PLAN

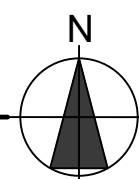
SHEET:

E200

JOB NUMBER:

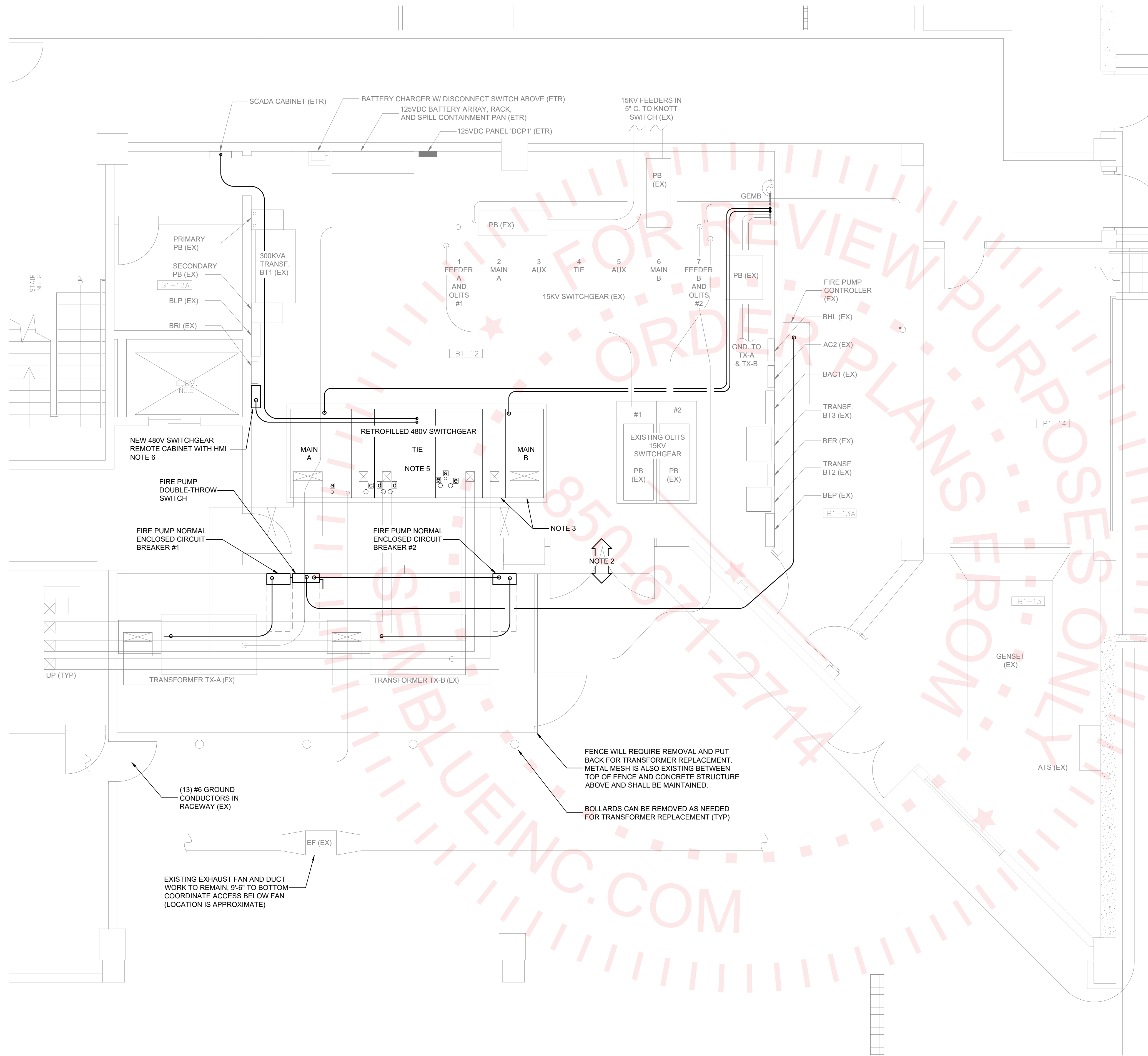
2311

DEMOLITION POWER PLAN
SCALE: 1/4"=1'-0"



UNDERGROUND FEEDERS IN 480V SWITCHGEAR

- Ⓐ GROUNDING ELECTRODE CONDUCTOR
- Ⓑ 3#4/0, 1#4 G., 2 1/2" C. TO FIRE PUMP CONTROLLER
- Ⓒ 4#500, 1#3 G., 3 1/2" C. TO PANEL 'BAC1'
- Ⓓ (2) 4#500, 1#2/0 G., 3 1/2" C. TO ATS
- Ⓔ (2) 3#250, 1#2 G., 2 1/2" C. TO TRANSF. BT1

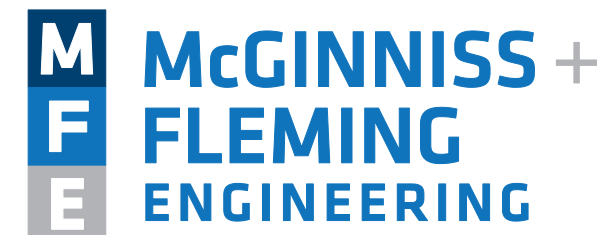


SHEET NOTES:

1. REFER TO SEQUENCING NOTES FOR PHASING OF EQUIPMENT REPLACEMENT.
2. EXISTING DOUBLE DOORS SHALL BE USED FOR ACCESS INTO ELECTRICAL ROOM FOR EQUIPMENT REPLACEMENT. ELECTRICAL ROOM FLOOR ELEVATION IS 48" ABOVE GARAGE FLOOR. DOOR OPENING IS APPROX. 7'-10" WIDE AND HAS A CLEAR HEAD HEIGHT OF APPROX. 8'-5". DOOR HAS REMOVABLE CONCRETE THRESHOLD TO ALLOW FOR EQUIPMENT PASS-THROUGH. CONTRACTOR SHALL REMOVE AS NEEDED AND PUT BACK WHEN COMPLETED WITH MOVING EQUIPMENT THROUGH.
3. TWO SECTIONS OF RETROFITTED 480V SWITCHGEAR THAT CANNOT MEET WORKING CLEARANCE REQUIREMENTS AT REAR PANELS DUE TO STRUCTURAL COLUMNS SHALL HAVE TAMPER-PROOF HARDWARE FOR SECURING BACK PANELS. REPLACE EXISTING HARDWARE, MATCH EXISTING SIZES. THESE PANELS WILL NOT BE PERMITTED TO BE OPENED WHILE ENERGIZED. PROVIDE SIGNAGE AND SUPPLY OWNER WITH (2) SECURITY TIPS TO MATCH HARDWARE PROVIDED.
4. NOTE CONFIGURATION OF DRY-TYPE TRANSFORMERS ARE PRIMARY ON RIGHT SIDE AND SECONDARY ON LEFT SIDE, FROM FRONT VIEW. FIRE PUMP TAPS SHALL OCCUR ON SECONDARY 480V BUS.
5. 480V SWITCHGEAR ENCLOSURE FRAME, FRONT, REAR, AND SIDE PANELS SHALL BE PROFESSIONALLY CLEANED AND PROFESSIONALLY PAINTED TO MATCH NEW 15KV SWITCHGEAR FINISH. NEW 15KV SWITCHGEAR IS SPECIFIED TO BE PAINTED MEDIUM GRAY ANSI #49. NEW BREAKER DOOR PANELS SHALL ALSO BE MEDIUM GRAY ANSI #49. REMOVE NAMEPLATES FROM GEAR PRIOR TO PAINTING, STORE AND SECURE, AND RE-INSTALL WHEN COMPLETED USING SCREWS OR RIVETS.
6. NEW WALL-MOUNT REMOTE PANEL SHALL INCLUDE HMI FOR SWITCHGEAR CONTROLS AS NOTED. PLC AND POWER SUPPLY SHALL BE INSTALLED WITHIN AUX. SECTION OF SWITCHGEAR. PROVIDE PROPERLY SIZED RACEWAY AND WIRING BETWEEN GEAR AND WALL-MOUNT HMI PANEL.

UNDERGROUND FEEDERS IN 480V SWITCHGEAR

- Ⓐ GROUNDING ELECTRODE CONDUCTOR
- Ⓑ NOT USED.
- Ⓒ 4#500, 1#3 G., 3 1/2" C. TO PANEL 'BAC1'
- Ⓓ (2) 4#500, 1#2/0 G., 3 1/2" C. TO ATS
- Ⓔ (2) 3#250, 1#2 G., 2 1/2" C. TO TRANSF. BT1



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PEPPER BUILDING
DESIGN REPLACEMENT OF
MAJOR ELECTRICAL SYSTEMS
PHASE 2

MSFM-02203280

Tallahassee, Florida

DATE:
April 30, 2024

REVISED:

DESIGNED BY:
BW

DRAWN BY:
TB

SUBMITTAL:
100% Construction Documents

SHEET TITLE:

NEW WORK
POWER PLAN

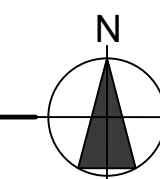
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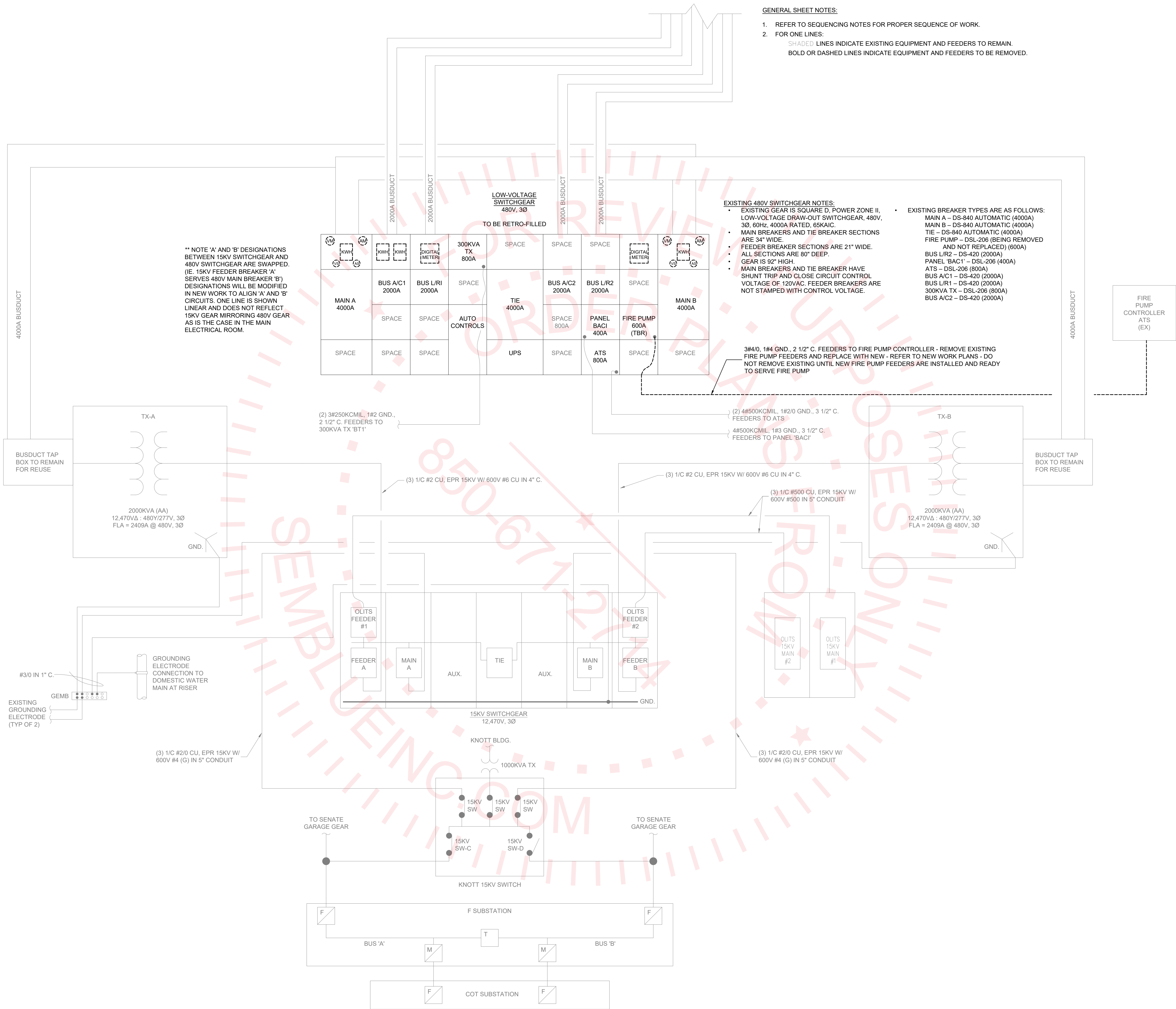
E300

JOB NUMBER:

2311

NEW WORK POWER PLAN
SCALE: 1/4"=1'-0"





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MAJOR ELECTRICAL SYSTEMS
PHASE 2

MSFM-02203280

Tallahassee, Florida

DATE:
April 30, 2024

REVISED:

DESIGNED BY:
BW

DRAWN BY:
TB

SUBMITTAL:
100% Construction Documents

SHEET TITLE:

EXISTING
ONE LINE DIAGRAM

SHEET:

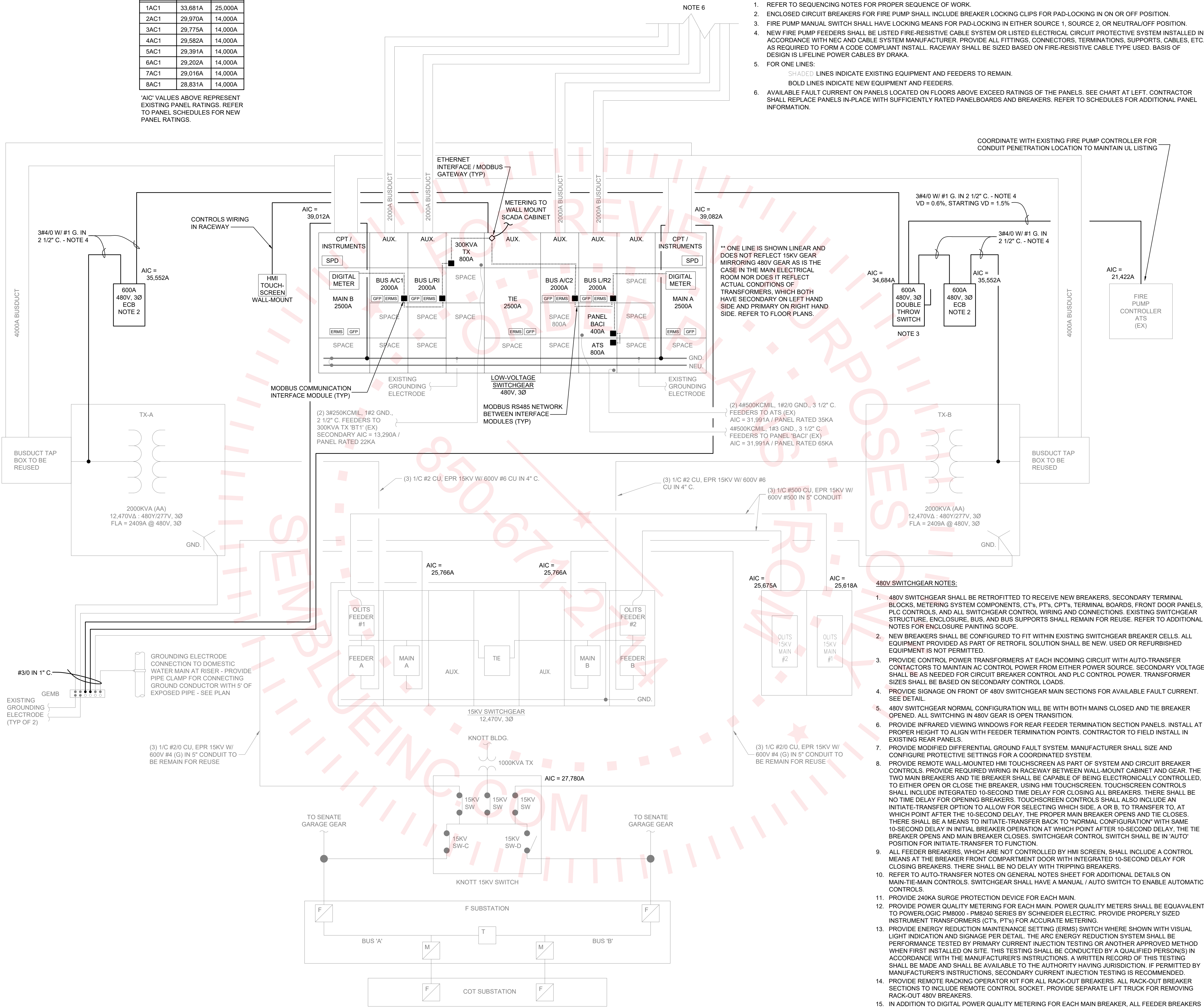
E400

JOB NUMBER:

2311

PANEL	AFC	AIC
1AC1	33,681A	25,000A
2AC1	29,970A	14,000A
3AC1	29,775A	14,000A
4AC1	29,582A	14,000A
5AC1	29,391A	14,000A
6AC1	29,202A	14,000A
7AC1	29,016A	14,000A
8AC1	28,831A	14,000A

'AIC' VALUES ABOVE REPRESENT EXISTING PANEL RATINGS. REFER TO PANEL SCHEDULES FOR NEW PANEL RATINGS.



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PEPPER BUILDING
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MAJOR ELECTRICAL SYSTEMS
PHASE 2

MSFM-02203280

Tallahassee, Florida

DATE:

April 30, 2024

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100% Construction Documents

SHEET TITLE:

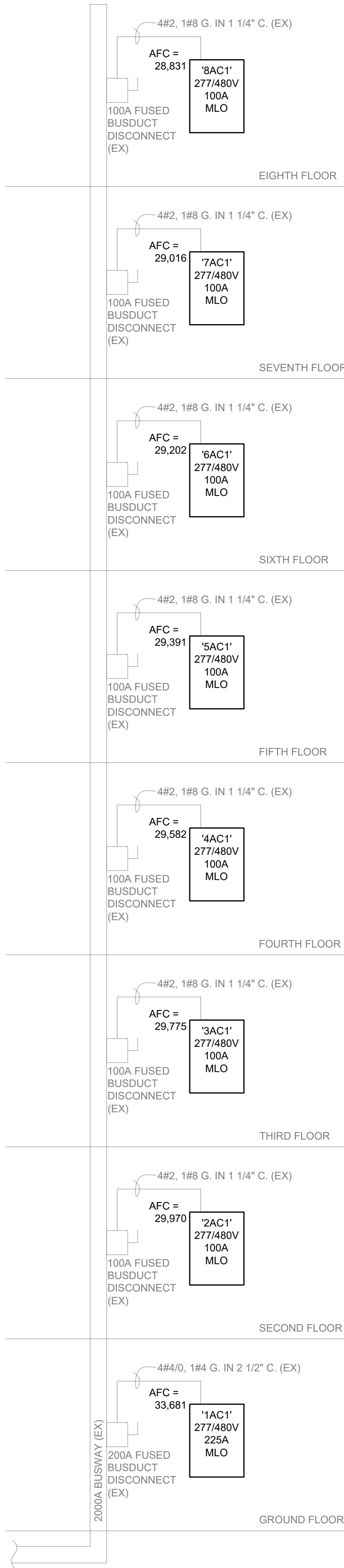
NEW
ONE LINE DIAGRAM

SHEET:

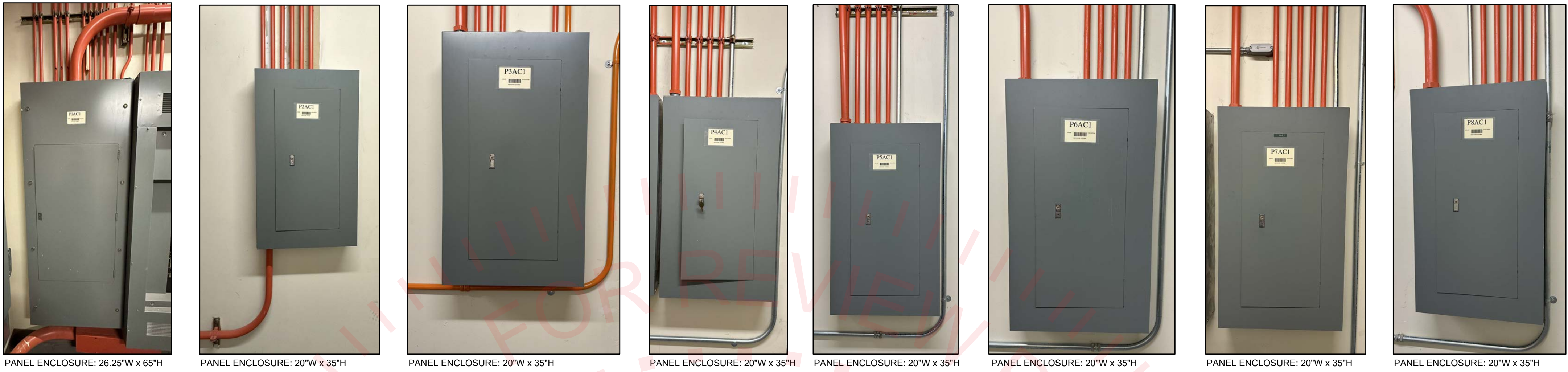
E401

JOB NUMBER:

2311



PARTIAL BUSDUCT A/C2 ONE LINE
NOT TO SCALE



PANEL '1AC1'
NOT TO SCALE

PANEL '2AC1'
NOT TO SCALE

PANEL '3AC1'
NOT TO SCALE

PANEL '4AC1'
NOT TO SCALE

PANEL '5AC1'
NOT TO SCALE

PANEL '6AC1'
NOT TO SCALE

PANEL '7AC1'
NOT TO SCALE

PANEL '8AC1'
NOT TO SCALE

LOAD / REMARKS	CIR. NO.	BREAKERS AMPS POLES	H.P. OR K.W.	LIGHTING			AMPS PER PHASE			SPECIAL			H.P. OR K.W.	BREAKERS AMPS POLES	CIR. NO.	LOAD / REMARKS
				ØA	ØB	ØC	ØA	ØB	ØC	ØA	ØB	ØC				
FTU - EAST SIDE	1	20	3	-	-	-	-	-	-	-	-	-	-	20	3	2
	3			-	-	-	-	-	-	-	-	-	-			4
	5			-	-	-	-	-	-	-	-	-	-			6
AHU G-5	7	20	3	-	-	-	-	-	-	-	-	-	-	20	3	8
	9			-	-	-	-	-	-	-	-	-	-			10
	11			-	-	-	-	-	-	-	-	-	-			12
SPACE	13			-	-	-	-	-	-	-	-	-	-			14
SPACE	15			-	-	-	-	-	-	-	-	-	-			16
SPACE	17			-	-	-	-	-	-	-	-	-	-			18
FTU-7	19	25	1	-	-	-	-	-	-	-	-	-	-	25	1	20
FTU-8	21	25	1	-	-	-	-	-	-	-	-	-	-	25	1	22
FTU-9	23	25	1	-	-	-	-	-	-	-	-	-	-	25	1	24
FTU-10	25	25	1	-	-	-	-	-	-	-	-	-	-	25	1	26
FTU-11	27	25	1	-	-	-	-	-	-	-	-	-	-	25	1	28
FTU-12	29	25	1	-	-	-	-	-	-	-	-	-	-	25	1	30
EST. CONNECTED LOAD (AMPS)														EST. CONNECTED LOAD (AMPS)		

PANELBOARD SCHEDULE

PEPPER BUILDING
Tallahassee, Florida

Panel 1AC1

Location ELECTRICAL RM.

Fed From BUS DUCT

Service 3 Phase 4 Wire
☐ 208Y/120V
☒ 480Y/277V

Main Bkr. - A - P

Lugs Only 225 A

42,000 AIC Min.
60 Hz.

☒ Surface mounted panel

SHEET NOTES:

- PANELS SHOWN SHALL BE REPLACED IN-PLACE. CONTRACTOR SHALL DISCONNECT EXISTING FEEDERS AND CIRCUITS. PULL BACK AS REQUIRED TO REWORK CONDUIT AND FEEDERS INTO NEW PANELBOARD.
- LABEL ALL WIRING PRIOR TO DISCONNECTING FROM EXISTING BREAKERS TO MATCH UP WITH PROPER BREAKER IN NEW PANEL.
- BUSWAY SWITCHES AND FEEDERS ARE EXISTING TO REMAIN.
- PHOTOS SHOWN ARE INTENDED TO GIVE CONTRACTOR AN IDEA OF WHAT CONDUIT ENTERS / EXITS THE PANEL AND WILL REQUIRE REWORKING. CONTRACTOR TO COORDINATE WITH ONSITE CONDITIONS AND EXISTING PANEL CONFIGURATIONS FOR FURTHER DETAILS. DIMENSIONS NOTED ARE OF EXISTING PANELBOARD ENCLOSURES.
- PROVIDE NEW NAMEPLATES AND TYPEWRITTEN SCHEDULES FOR ALL NEW PANELS.

LOAD / REMARKS	CIR. NO.	BREAKERS AMPS POLES	H.P. OR K.W.	LIGHTING			AMPS PER PHASE			SPECIAL			H.P. OR K.W.	BREAKERS AMPS POLES	CIR. NO.	LOAD / REMARKS
				ØA	ØB	ØC	ØA	ØB	ØC	ØA	ØB	ØC				
SPARE	1	20	1	-	-	-	-	-	-	-	-	-	-	25	1	2
SPACE	3			-	-	-	-	-	-	-	-	-	-	25	1	4
SPACE	5			-	-	-	-	-	-	-	-	-	-	25	1	6
SPACE	7			-	-	-	-	-	-	-	-	-	-	25	1	8
SPACE	9	20	1	-	-	-	-	-	-	-	-	-	-	25	1	10
SPACE	11	20	1	-	-	-	-	-	-	-	-	-	-	25	1	12
SPACE	13	20	1	-	-	-	-	-	-	-	-	-	-	25	1	14
SPACE	15	20	1	-	-	-	-	-	-	-	-	-	-	25	1	16
SPACE	17	20	1	-	-	-	-	-	-	-	-	-	-	25	1	18
SPACE	19	20	1	-	-	-	-	-	-	-	-	-	-	25	1	20
SPACE	21	20	1	-	-	-	-	-	-	-	-	-	-	25	1	22
SPACE	23	20	1	-	-	-	-	-	-	-	-	-	-	25	1	24
SPACE	25	20	1	-	-	-	-	-	-	-	-	-	-	25	1	26
SPACE	27	20	1	-	-	-	-	-	-	-	-	-	-	25	1	28
SPACE	29	20	1	-	-	-	-	-	-	-	-	-	-	20	1	30
EST. CONNECTED LOAD (AMPS)														EST. CONNECTED LOAD (AMPS)		

PANELBOARD SCHEDULE

PEPPER BUILDING
Tallahassee, Florida

Panel 2AC1

Location ELECTRICAL RM.

Fed From BUS DUCT

Service 3 Phase 4 Wire
☐ 208Y/120V
☒ 480Y/277V

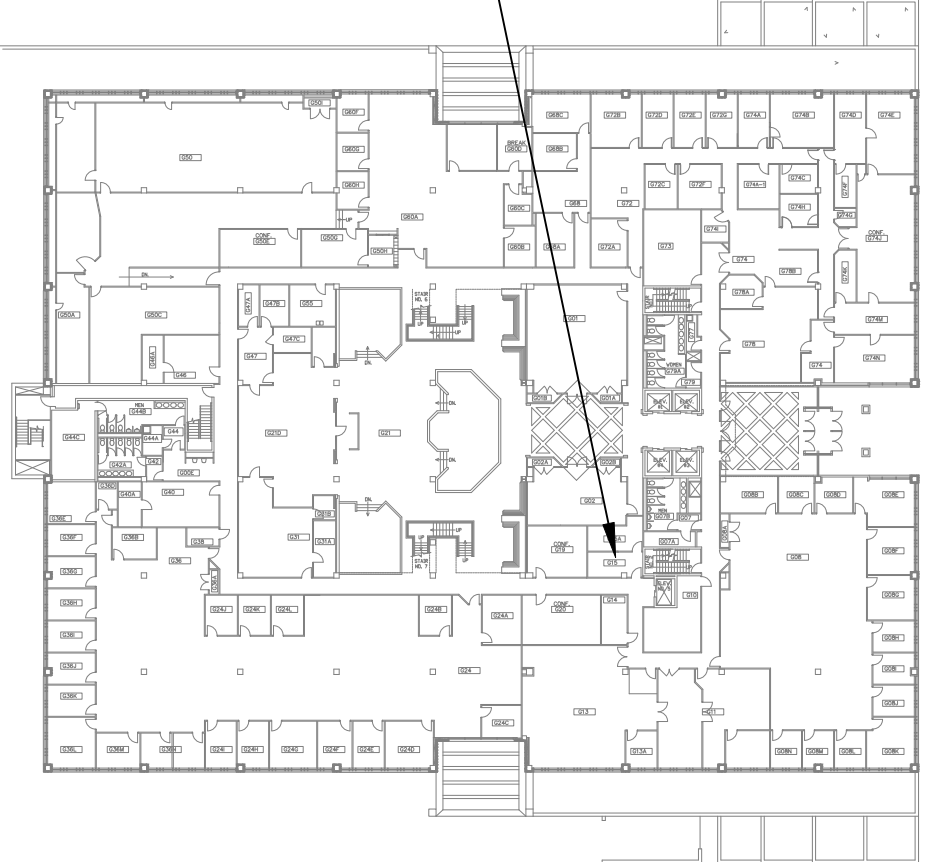
Main Bkr. - A - P

Lugs Only 100 A

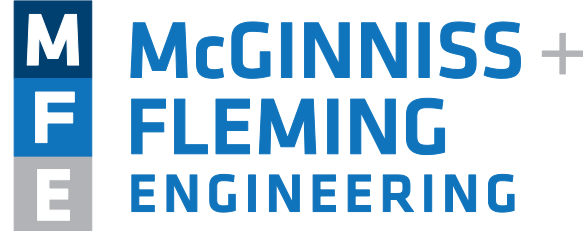
42,000 AIC Min.
60 Hz.

☒ Surface mounted panel

STACKED ELECTRICAL
ROOM LOCATIONS
(TYP EACH FLOOR)



GROUND FLOOR KEYPLAN
NOT TO SCALE



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PEPPER BUILDING
DESIGN REPLACEMENT OF
MAJOR ELECTRICAL SYSTEMS
PHASE 2

MSFM-02203280

Tallahassee, Florida

DATE:
April 30, 2024

REVISED:

DESIGNED BY:
BW

DRAWN BY:
TB

SUBMITTAL:
100% Construction Documents

SHEET TITLE:

PARTIAL ONE LINE, PHOTOS, AND
PANEL SCHEDULES

SHEET:

E402

JOB NUMBER:

2311

PANELBOARD SCHEDULE	
PEPPER BUILDING	
Tallahassee, Florida	
Panel	<u>3AC1</u>
Location	<u>ELECTRICAL RM.</u>
Fed From	<u>BUS DUCT</u>
Service	<u>3</u> Phase <u>4</u> Wire
☐ 208Y/120V	
☒ 480Y/277V	
Main Bkr.	<u>-</u> A <u>-</u> P
Lugs Only	<u>100</u> A
<u>42,000</u> AIC Min.	
<u>60</u> Hz.	
☒ Surface mounted panel	

PANELBOARD SCHEDULE

PEPPER BUILDING
Tallahassee, Florida

Panel 4AC1

Location ELECTRICAL RM.

Fed From BUS DUCT

Service 3 Phase 4 Wire

☐ 208Y/120V

☒ 480Y/277V

Main Bkr. - A - P

Lugs Only 100 A

42,000 AIC Min.

60 Hz.

☒ Surface mounted panel

PANELBOARD SCHEDULE	
PEPPER BUILDING Tallahassee, Florida	
Panel	<u>5AC1</u>
Location	<u>ELECTRICAL RM.</u>
Fed From	<u>BUS DUCT</u>
Service	<u>3</u> Phase <u>4</u> Wire
☐ 208Y/120V ☒ 480Y/277V	
Main Bkr.	<u>-</u> A <u>-</u> P
Lugs Only	<u>100</u> A
<u>42,000</u> AIC Min. <u>60</u> Hz.	
☒ Surface mounted panel	

PANELBOARD SCHEDULE

PEPPER BUILDING
Tallahassee, Florida

Panel 6AC1

Location ELECTRICAL RM.

Fed From BUS DUCT

Service 3 Phase 4 Wire
☐ 208Y/120V
☒ 480Y/277V

Main Bkr. - A - P

Lugs Only 100 A

42,000 AIC Min.
60 Hz.

☒ Surface mounted panel

PANELBOARD SCHEDULE			
PEPPER BUILDING			
Tallahassee, Florida			
Panel	7AC1		
Location	ELECTRICAL RM.		
Fed From	BUS DUCT		
Service	3	Phase	4 Wire
☐ 208Y/120V ☒ 480Y/277V			
Main Bkr.	- A	-	P
Lugs Only	100	A	
42,000	AIC Min.		
60	Hz.		
☒ Surface mounted panel			

PANELBOARD SCHEDULE

PEPPER BUILDING
Tallahassee, Florida

Panel	<u>8AC1</u>
Location	<u>ELECTRICAL RM.</u>
Fed From	<u>BUS DUCT</u>
Service	<u>3</u> Phase <u>4</u> Wire
☐ 208Y/120V	
■ 480Y/277V	
Main Bkr.	<u>-</u> <u>A</u> <u>-</u> <u>P</u>
Lugs Only	<u>100</u> <u>A</u>
<u>42,000</u> AIC Min.	
<u>60</u> Hz.	
■ Surface mounted panel	



PEPPER BUILDING
DESIGN REPLACEMENT OF
MAJOR ELECTRICAL SYSTEMS
PHASE 2
MSFM-02203280
Tallahassee, Florida

REVISÉ:

DRAWN BY:
TB

SHEET TITLE:
PANEL SCHEDULES

SHEET:

E403

JOB NUMBER:

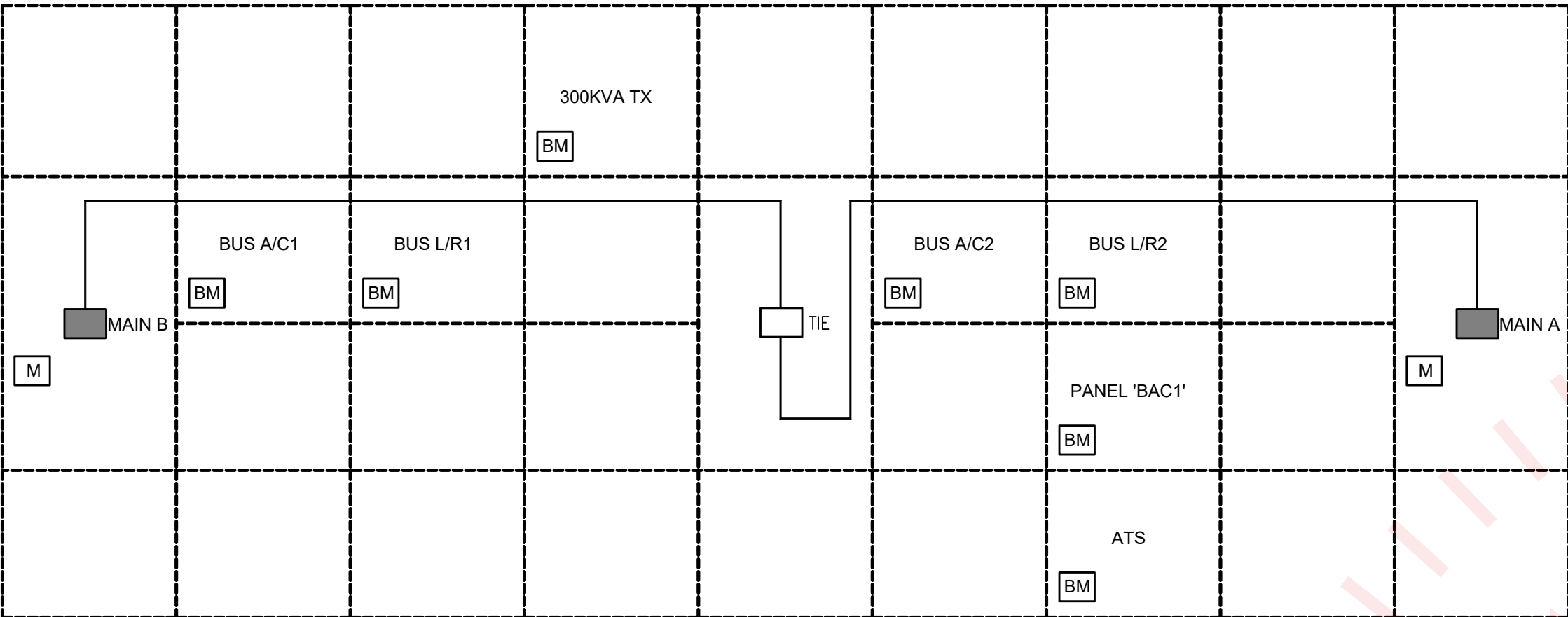
2311

SCADA SYSTEM GRAPHICS

DATA (SCADA) NOTES

SCADA NETWORK ONE LINE

480V SWITCHBOARD



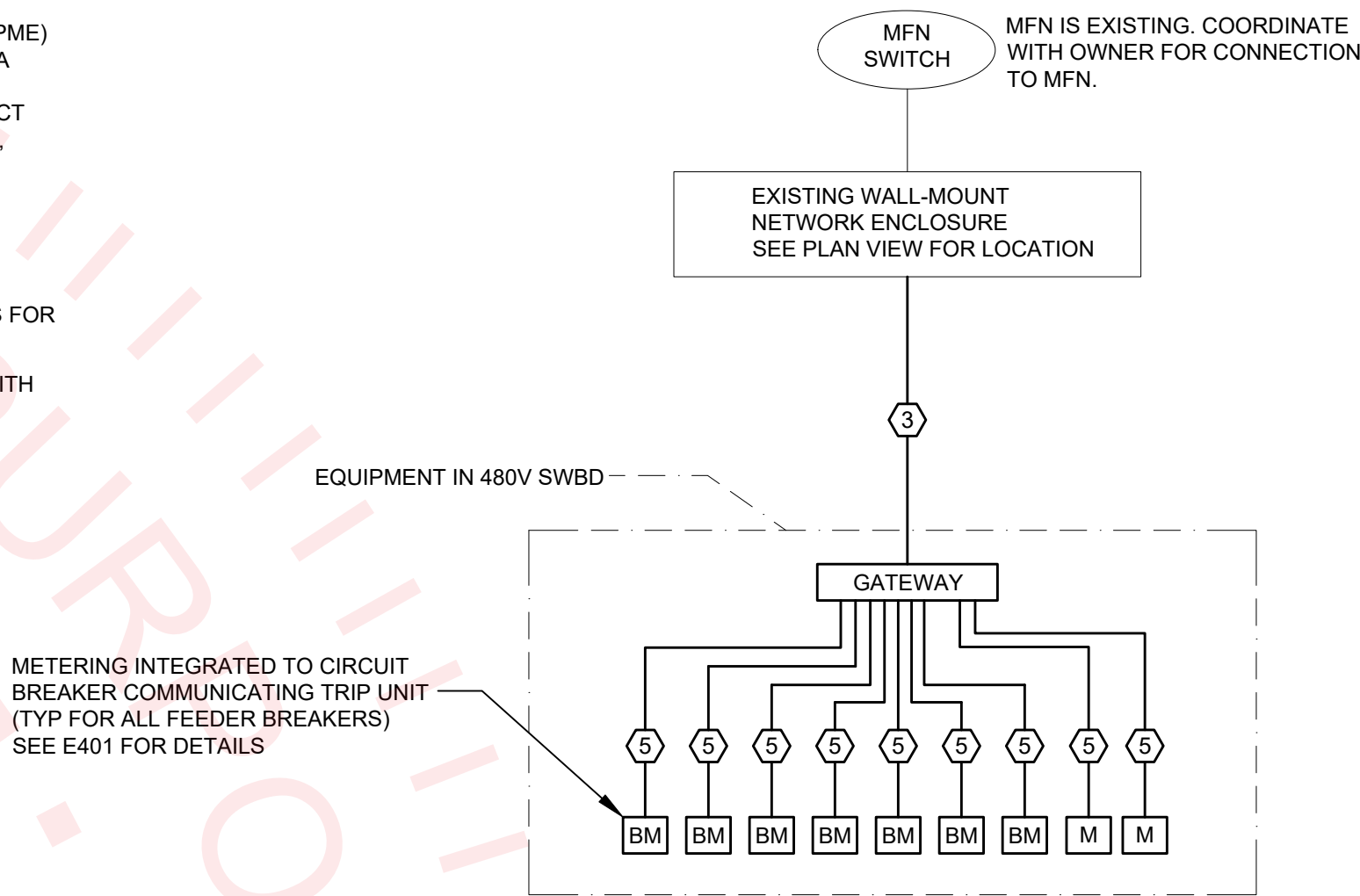
NOTE: 480V SWGR SHOWN HERE IS DIAGRAMMATIC. PME SHALL BE UPDATED TO INCLUDE DRAWINGS FOR EACH COMPONENT WITHIN THE EQUIPMENT NOTED TO INTERACT WITH PME SYSTEM.

- M DIGITAL METERING
- BM BREAKER ELECTRONIC METERING
- CIRCUIT BREAKER STATUS - CLOSED
- CIRCUIT BREAKER STATUS - OPEN

- SCADA SHALL INTERACT WITH NEW EQUIPMENT COMPONENTS FOR THE ACTIONS NOTED.
- FOR 480V SWGR, SCADA SHALL MONITOR TWO MAINS AND TIE BREAKER AND INDICATE GRAPHICALLY THE STATUS OF EACH - OPEN OR CLOSED. DEMONSTRATE PROPER SIGNALING AT COMPLETION OF PROJECT.
- SCADA SHALL MONITOR DEDICATED MAIN BREAKER POWER QUALITY DIGITAL METERING IN 480V SWGR TO PROVIDE FULL METERING INFORMATION TO SCADA SOFTWARE.
- SCADA SHALL MONITOR FEEDER BREAKER ELECTRONIC TRIP UNITS IN 480V SWGR TO PROVIDE FULL METERING INFORMATION TO SCADA SOFTWARE.
- SCADA SYSTEM SHALL INCLUDE GRAPHIC OF 480V SWGR SIMILAR TO GRAPHIC SHOWN AT LEFT WITH INTERACTIVE INTERFACE.
- ALL BREAKERS AND METERING EQUIPMENT SHALL BE CONFIGURED AND PROGRAMMED ACCORDINGLY TO MEET NOTED REQUIREMENTS FOR PROPER SCADA CONFIGURATION.
- NO REMOTE CONTROL FEATURES OF SWITCHGEAR SHALL BE CONFIGURED WITH SCADA SYSTEM.
- EXISTING SCADA SYSTEM IS ECOSTRUXURE POWER MONITORING EXPERT (PME) BY SCHNEIDER ELECTRIC. CONTRACTOR SHALL SECURE THE SERVICES OF A CERTIFIED TECHNICIAN EXPERIENCED WITH PME SOFTWARE TO PERFORM CHANGES TO THE EXISTING SCADA SYSTEM TO ACCOMMODATE THE PROJECT REQUIREMENTS. CHANGES INCLUDE ALL NECESSARY SOFTWARE, UPDATES, LICENSES, HARDWARE COMPONENTS, PROGRAMMING OF NEW DEVICES, REMOVAL OF EXISTING DEVICES NOTED TO BE REMOVED FROM SYSTEM, GRAPHIC DEVELOPMENT, AND OWNER TRAINING.
- EQUIPMENT PROVIDED, REGARDLESS OF MANUFACTURER, SHALL BE CONFIGURED FOR A FULLY INTEGRATED SYSTEM AS DETAILED ABOVE. CONTRACTOR SHALL GUARANTEE THAT ALL EQUIPMENT PROVIDED ALLOWS FOR FULL INTEGRATION INTO THE OWNER'S PME SYSTEM AS NOTED.
- MANUFACTURER SHALL INCLUDE QUANTITY OF IP ADDRESSES REQUIRED WITH SUBMITTAL PACKAGE FOR OWNER'S REVIEW AND PLANNING.

NETWORK EQUIPMENT SCHEDULE:

- 2-STRAND, 62.5/125, MM OPTICAL FIBER BY SWITCHGEAR MANUFACTURER
- 6-STRAND, 62.5/125, MM OPTICAL FIBER IN 1" CONDUIT, UNO (COLOR ORANGE)
- UNSHIELDED TWISTED PAIR, CAT 6 ETHERNET CABLE, PLENUM RATED, 24 GAUGE, SOLID BARE ANNEALED COPPER, 10Gb/s, FEP INSULATION, TERMINATED EACH END. (COLOR ORANGE)
- HARDENED, MANAGED, FANLESS DESIGN, INDUSTRIAL NETWORK FIBER SWITCH WITH 1000BASE-SX 850NM MODULE(S) (GUIDE: HIRSCHMANN)
- SWGR INTERNAL WIRING GATEWAY TO SINGLE ETHERNET PORT IN SWGR - SEE E401



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PEPPER BUILDING
DESIGN REPLACEMENT OF
MAJOR ELECTRICAL SYSTEMS
PHASE 2

MSFM-02203280

Tallahassee, Florida

DATE:

April 30, 2024

REVISED:

DESIGNED BY:

BW

DRAWN BY:

TB

SUBMITTAL:

100% Construction Documents

SHEET TITLE:

SCADA PLAN
AND NOTES

SHEET:

E500

JOB NUMBER:

2311